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Physics-guided multi-fidelity framework for probabilistic deformation prediction of concrete arch dams under digital twin-based monitoring

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ABSTRACT

Digital twin-enabled safety monitoring of concrete arch dams requires deformation prediction models that are continuously updateable, physically informed, and uncertainty-aware. Nevertheless, existing approaches often struggle to effectively integrate simulation knowledge with monitoring data, especially under the dual constraints of limited response time and previously unseen real operating conditions. To overcome these challenges, this study proposes a physics-guided multi-fidelity Bayesian neural network (PGMF-BNN) framework for digital twin-oriented probabilistic deformation prediction. Low-fidelity data are generated from a three-dimensional finite element model to represent the dominant deformation behavior under hydrostatic loading initially. A multi-input multi-output (MIMO) architecture is constructed to capture spatial dependencies among monitoring points and enable collaborative prediction. The model is subsequently pre-trained using low-fidelity data to learn a mechanically consistent response skeleton, followed by a two-stage transfer learning process using prototype monitoring data to achieve virtual–physical alignment. Finally, Bayesian inference is incorporated to quantify epistemic and aleatoric uncertainties and to produce calibrated prediction intervals. The effectiveness of the proposed framework is demonstrated using a real concrete arch dam case. The results indicate that the proposed method achieves superior predictive performance, with coefficients of determination exceeding 0.90 at five of the six representative monitoring points (reaching 0.975 at PLA2) and mean absolute percentage errors below 7%. The model successfully reconstructs the spatial correlation structure of multi-point deformation and demonstrates robust performance under limited-data conditions and moderate noise levels (0%–10%). These findings confirm that the proposed framework delivers accurate, physically consistent, and uncertainty-aware deformation predictions under both normal and disturbed operating conditions, providing a reliable and extensible solution for digital twin-based dam safety monitoring and risk early warning.

1. Introduction

Concrete arch dams are critical hydraulic infrastructures characterized by high structural efficiency, strong adaptability to complex geological conditions, and long service lifetimes[1–4]. Ensuring their operational safety requires accurate characterization and prediction of structural responses under continuously evolving environmental and loading conditions[5–8]. In recent years, the emergence of digital twin technologies has fundamentally transformed dam safety management from conventional offline assessment to real-time and model-integrated monitoring paradigms[9–12]. Under this condition, deformation prediction is no longer limited to retrospective fitting, but is expected to support continuous real-time state awareness, trend inference, and early warning under nonstationary operating conditions[13,14].

Among the various monitoring indicators, deformation provides a direct and sensitive measure of the structural response to external loads. However, the deformation behavior of concrete arch dams is governed by the coupled effects of hydrostatic pressure, temperature variation, and time-dependent material properties, resulting in highly nonlinear, path-dependent, and temporally evolving characteristics[15,16]. These intrinsic complexities pose significant challenges to the development of predictive models that are simultaneously accurate, physically consistent, and robust under diverse operating scenarios.

Existing approaches for dam deformation prediction can be broadly

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classified into statistical models, deterministic numerical models, and data-driven methods. Statistical models are widely used due to their computational efficiency and interpretable parameterization of engineering factors. For example, Mata et al. [17] proposed a water level–temperature–time (HTT) statistical model based on principal component analysis (PCA), designed to replace the seasonal sine function in the traditional water level–season–time (HST) model with measured temperature data from within the dam body, thereby more accurately capturing the influence of thermal effects on the deformation of arch dams. Studies have shown that this model significantly outperforms traditional models in terms of displacement prediction accuracy and reduction of residual standard deviation. Sigtryggsdóttir et al. [18] developed an HST multi-source linear statistical model for predicting in-service settlement of concrete-faced rock-fill dams (CFRDs), correlating each statistical component with a viscoelastic-plastic constitutive model. The model innovatively accounts for unloading–reloading behavior caused by water level fluctuations and the creep effects of rock fill. These studies demonstrate how empirical monitoring knowledge can be formalized through physically meaningful statistical components. Nevertheless, their reliance on predefined functional forms and linear assumptions limits their capability in representing complex nonlinear interactions among influencing factors [19]. To address this limitation, some researchers have opted to employ deterministic numerical simulation models, such as the finite element method, which offers an alternative formalized approach to engineering knowledge by directly embedding structural mechanics and boundary conditions into the prediction process. For example, Xiong et al.[20] built upon traditional deterministic displacement models by introducing a residual sequence correction approach. They first decomposed the residuals using CEEMDAN and multiscale permutation entropy; the high-frequency components were denoised using wavelets and then predicted by a GRU-LSTM, while the low-frequency components were predicted by an ARIMA model. Finally, these results were combined with those from the deterministic model to form a hybrid prediction model. Results demonstrate that this model better captures the chaotic and uncertain information within the residuals, thereby improving the prediction accuracy of arch dam displacement. Yang et al.[21] addressed the shortcoming of traditional HST models in adequately characterizing the combined effects of "structure-topography-geology-high water pressure" in high-arch dams. They proposed a hybrid HST model that couples water pressure-induced deformation obtained from elastic three-dimensional finite element simulations with seasonal and temporal terms. They performed full-parameter optimization using an improved PSO algorithm that incorporates stochastic inertia weights and dynamic learning factors. Case studies demonstrate that this model achieves higher fitting and prediction accuracy compared to traditional models. These studies indicate that numerical simulations can strengthen the mechanical interpretability of dam response prediction. However, their practical application in real-time prediction is constrained by high computational cost, model simplifications, and uncertainties in material parameters and boundary conditions, which often lead to systematic discrepancies between simulated and measured responses[22,23].

To further enhance the characterization ability of complex nonlinear responses of dams and improve the shortcomings of traditional models in real-time performance, data-driven models represented by deep learning have gradually become an important research direction for dam safety prediction. This type of model, with its excellent feature extraction ability, can efficiently and accurately capture the nonlinear features of structural deformation. For example, Jia et al.[24] used a PSO-optimized XGBoost model and SHAP values to analyze the contributions of water pressure, temperature, and time-effect factors, thereby achieving mechanistic clustering of monitoring points. Based on this, they employed a PSO-optimized iTransformer to construct a panel data prediction model capable of capturing spatiotemporal correlations across multiple monitoring points. This method overcomes the shortcomings of traditional clustering that ignores factor contribution and single point prediction that cannot reflect overall deformation. Yang et al.[25] utilized an improved cloud model to extract digital features from displacement sequences and employed Agglomerative Hierarchical Clustering (AHC) to partition monitoring points with similar deformation patterns; subsequently, based on the partitioning results, they constructed a deep learning multi-input multi-output (MIMO) prediction framework that integrates CNN and BiLSTM to simultaneously predict the displacements of multiple monitoring points within each partition, while further enhancing model performance through batch normalization, dropout, and early stopping mechanisms, providing an efficient and practical new method for dam structural health monitoring. These studies show that deep learning can transform monitoring data into high-dimensional feature representations and capture spatially coordinated deformation patterns. However, purely data-driven approaches still suffer from several critical limitations in engineering practice. In particular, the scarcity and noise of monitoring data, the lack of extrapolation capability under unseen operating conditions, and the absence of physical constraints may result in predictions that violate fundamental mechanical principles, thereby undermining their reliability in safety–critical applications[26,27].

In response to these challenges, multi-fidelity surrogate modeling has emerged as a hot research topic in recent years. Multi-fidelity modeling seeks to balance physical laws and prediction accuracy by integrating low-fidelity numerical simulation data with high-fidelity experimental data. Its core concept involves using simulation data to construct a fundamental mechanical framework and then applying nonlinear corrections based on experimental data sequences[28]. For example, Guo et al.[29] used spatial mapping techniques to integrate numerical simulations from low-fidelity extended finite element models (XFEM) with high-fidelity in-situ monitoring data. By employing multi-fidelity surrogate models to establish complex mappings between damage parameters and structural responses in their predictions, they significantly improved identification accuracy and computational efficiency while avoiding repetitive grid division. Multi-fidelity approaches provide an effective pathway toward balancing physical interpretability and predictive accuracy[30]. However, when applied to complex dam systems, several key challenges remain unresolved. First, the discrepancy between virtual simulation models and real structural responses is difficult to eliminate, particularly under time-varying operational conditions. Second, most existing studies focus on single-point prediction and fail to explicitly model the spatial coupling and collaborative behavior of deformation across the dam body. Third, uncertainty quantification is often treated in a simplified manner, without distinguishing between epistemic uncertainty arising from model structure and parameter limitations, and aleatoric uncertainty induced by environmental variability and measurement noise. These limitations hinder the construction of reliable probabilistic prediction boundaries and restrict the applicability of existing methods in digital twin-based real-time monitoring and risk assessment.

To address the above challenges, this study proposes a physics-guided multi-fidelity Bayesian neural network (PGMF-BNN) framework for probabilistic deformation prediction of concrete arch dams in a digital twin-based monitoring. The proposed framework integrates physical simulation, monitoring data, and probabilistic learning within a unified modeling architecture. Specifically, low-fidelity data generated from a three-dimensional finite element model are utilized to guide the pre-training of the neural network, enabling it to learn the fundamental mechanical response patterns of the dam. A two-stage transfer learning strategy is then introduced to systematically reduce the discrepancy between simulated and measured responses under operational conditions. Furthermore, a multi-input multi-output (MIMO) architecture is constructed to explicitly capture the spatial coupling characteristics among multiple monitoring points. To enhance prediction reliability, Bayesian inference combined with MC Dropout is employed to jointly quantify epistemic and aleatoric uncertainties,

thereby providing calibrated probabilistic prediction intervals for digital twin applications.

It should be pointed out that the novelty of this study does not lie in the separate use of multi-fidelity learning, transfer learning, MIMO modeling, or Bayesian inference. On the contrary, its main contribution lies in integrating these technologies into a digital twin framework guided by dam mechanisms. In this framework, low-fidelity finite element simulations are used to understand mechanical response skeleton dominated by hydrostatic pressure, high-fidelity monitoring data are used to correct residual responses related to temperature and time-effect, MIMO modeling is used to represent spatially coordinated deformations, and Bayesian inference is used to construct probabilistic monitoring boundaries.

The main contributions of this study are summarized as follows:

- A physics-guided multi-fidelity learning framework is developed to integrate finite element simulation data with monitoring data, enabling consistent representation of dam deformation under evolving operating conditions.
- A two-stage transfer learning strategy is proposed to bridge the gap between virtual simulation models and real structural responses, improving model adaptability and accuracy.
- A Bayesian neural network with MC Dropout is introduced to achieve unified quantification of epistemic and aleatoric uncertainties, enhancing the credibility of probabilistic prediction.
- A MIMO-based collaborative prediction architecture is established to capture spatial coupling relationships among monitoring points and improve multi-point prediction performance.

Unlike purely data-driven approaches, the proposed framework leverages multi-fidelity modeling to construct a mechanically consistent response skeleton with full operating-condition coverage, which is subsequently refined using monitoring data to enable real-time adaptive prediction under evolving and previously unseen operating conditions.

The remainder of this paper is organized as follows. Section 2 presents the overall research framework of the proposed digital twin-oriented deformation prediction approach. Section 3 describes the methodology, including deformation mechanism decomposition, the physics-guided multi-fidelity modeling framework, the multi-input multi-output (MIMO) architecture for spatial collaborative prediction, the two-stage transfer learning strategy for virtual–physical alignment, and the Bayesian neural network formulation with uncertainty quantification based on MC Dropout. Section 4 validates the proposed method through a real engineering case, with emphasis on predictive performance under normal operating conditions, ablation analysis, spatial collaborative verification across multiple monitoring points, physical consistency interpretation, probabilistic reliability assessment, and robustness evaluation under non-ideal operating conditions. Ultimately, Section 5 summarizes the main conclusions and outlines potential directions for future research.

2. Research framework

To address the challenges of virtual–physical inconsistency, adaptive monitoring under evolving operating conditions, and the lack of unified representation of multi-point structural responses in digital twin applications of concrete arch dams, a physics-guided multi-fidelity Bayesian neural network (PGMF-BNN) framework is developed for online state monitoring. As illustrated in Fig. 1, the proposed framework integrates physical simulation, monitoring data, and probabilistic learning within a unified digital twin paradigm. It follows a closed-loop modeling mechanism that combines multi-source data fusion, physics-informed initialization, data-driven correction, uncertainty-aware prediction, and spatial collaborative perception. Through this integration, the framework enables continuous alignment between virtual models and real structural responses, while dynamically updating monitoring benchmarks under changing operational conditions.

Specifically, the framework consists of five key components:

(1) Multi-source heterogeneous data integration

Low-fidelity simulation data generated from finite element models and high-fidelity field monitoring data are integrated within a unified data representation space. This integration establishes a transferable mapping between virtual simulations and real structural observations, providing the foundation for multi-fidelity learning and digital twin updating.

(2) Physics-guided initialization of the virtual mechanical representation

A physics-guided pre-training strategy is employed using low-fidelity simulation data to encode the dominant mechanical behavior of the dam, particularly the deformation response driven by hydrostatic loading. This process embeds structural mechanics knowledge into the model initialization, forming a physically consistent baseline representation for subsequent data-driven refinement.

(3) Virtual–physical discrepancy correction via residual learning

To bridge the gap between simulated and observed responses, high-fidelity monitoring data are incorporated through a residual learning mechanism. This component captures complex effects that are difficult to model explicitly, such as temperature variation and time-dependent behavior, enabling the digital twin model to evolve from a static simulation surrogate to a dynamically updated representation of the real

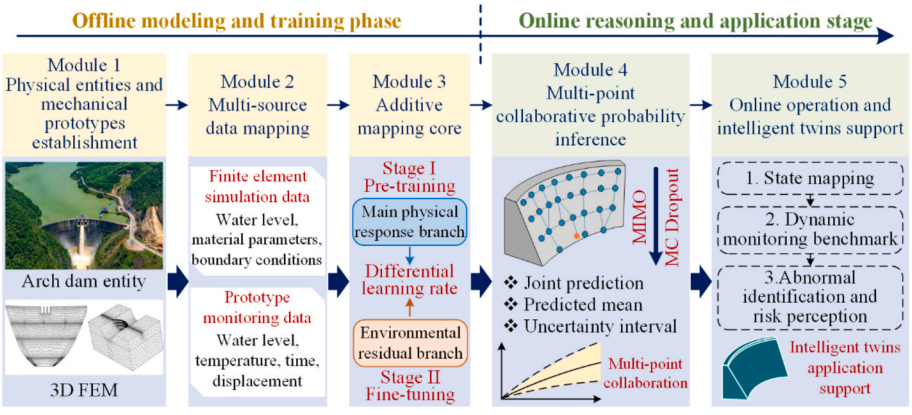


Fig. 1. Diagram of research framework.

structure.

(4) Probabilistic boundary construction and reliability quantification

To enhance prediction reliability, the framework extends deterministic outputs to probabilistic representations by introducing Bayesian inference. By modeling network parameters as random variables, both epistemic and aleatoric uncertainties are jointly quantified. The resulting probabilistic prediction intervals serve as dynamic health envelopes of the structure, providing a quantitative basis for anomaly detection and risk-aware decision-making.

(5) Multi-point collaborative state perception

Considering the strong spatial coupling characteristics of arch dam deformation, a multi-input multi-output (MIMO) architecture is adopted to jointly model multiple monitoring points. Through shared latent representations, the framework captures spatial correlation and collaborative response patterns across different dam regions, enabling system-level state perception beyond isolated single-point predictions.

Overall, the proposed framework establishes a scalable methodological foundation for intelligent monitoring and risk early warning of large-scale hydraulic structures.

3. Methodology

This section presents the methodology of the proposed physics-guided multi-fidelity digital twin framework for dam deformation prediction. A physically interpretable decomposition of deformation mechanisms is first introduced to establish the basis for model development. Building on this foundation, a physics-guided multi-fidelity framework is constructed, in which two-stage transfer learning is used for virtual–physical alignment and MIMO-based modeling is adopted to characterize spatial collaborative responses. The deterministic formulation is further extended to probabilistic prediction and uncertainty quantification by incorporating Bayesian modeling and MC Dropout. Finally, the implementation details are provided.

3.1. Decomposition mechanisms of deformation of concrete arch dams

Due to the coupled effects of multiple factors, such as reservoir water level fluctuations and cyclic temperature variations, the radial displacement of a concrete arch dam is difficult to characterize accurately using conventional mathematical formulations. To establish a physically interpretable basis for subsequent multi-fidelity modeling, the dam displacement response at time t is decomposed into the superposition of the hydrostatic-pressure component, the temperature-induced component, the time-effect component, and a random disturbance term, as illustrated in Fig. 2. This decomposition is consistent with the conventional understanding of water level, temperature, and time effects in dam safety monitoring[31,32], while also providing a structured physical basis for the subsequent learning of the dominant response and residual correction within the proposed dual-branch framework.

$$y(t) = y_H(t) + y_T(t) + y_C(t) + \varepsilon(t) \tag{1}$$

where $y(t)$ represents the total displacement response at time t. For multi-point monitoring, $y(t)$ can be extended from a scalar quantity to a displacement vector. Here, $y_H(t)$ denotes the displacement component caused by hydrostatic pressure; $y_T(t)$ represents the displacement component caused by temperature; $y_C(t)$ corresponds to the time-effect component associated with long-term material aging and rheological evolution. To better characterize thermal inertia, both the current temperature and the moving-average temperature are introduced in the subsequent modeling process. In addition, considering the long-term creep behavior of dam materials during operation, the logarithmic time factor $\ln(t+1)$ is adopted to describe the time effect, where t denotes the elapsed monitoring time interval from the initial monitoring date. This definition avoids the singularity of $\ln t$ at $t = 0$ and is consistent with the long-term convergent evolution of dam deformation. $\varepsilon(t)$ represents observation noise and other random errors.

Based on the above decomposition, the total displacement response is incorporated into a dual-branch learning architecture with explicitly differentiated functional roles. The physical branch is intended to learn the dominant hydrostatic-pressure-induced response and thereby establish the fundamental mechanical representation of dam deformation. In contrast, the residual branch focuses on modeling the deformation components associated with temperature variation and time-dependent effects, while simultaneously correcting the mismatch between the physics-based response and the actual monitored response. Furthermore, to enable a unified formulation for the multi-point collaborative prediction task, the scalar displacement response at a single monitoring point is extended to a multi-output displacement

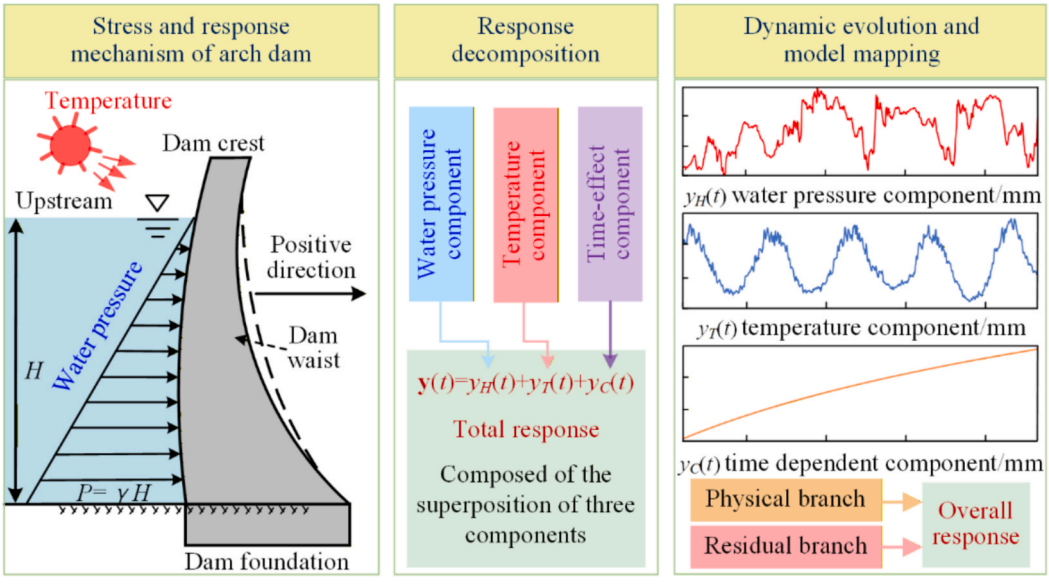


Fig. 2. Decomposition of deformation components and structural degradation mechanism.

vector corresponding to the ten monitoring points considered in this study:

$$\boldsymbol{y}(t) \in \mathbb{R}^N \tag{2}$$

where N denotes the number of monitoring points.

3.2. Physics-guided multi-fidelity MIMO framework of deformation prediction for concrete arch dams

3.2.1. Overall framework of the physics-guided multi-fidelity model

Conventional deterministic finite element models provide a well-established mechanical basis for describing dam behavior. However, within the context of dam safety monitoring and digital twin modeling, they are more appropriately treated as low-fidelity data sources. Here, the term "low-fidelity" (LF) does not imply that the numerical solutions themselves are inaccurate. Rather, it indicates that a non-negligible discrepancy may exist between simulation-based responses and the actual structural behavior, owing to uncertainties in material parameter identification, simplifications of boundary conditions, approximations in time-dependent constitutive modeling, and the omission of complex environmental disturbances[33]. Therefore, finite element models are more suitable for providing mechanistic prior knowledge and virtual prototypes within a digital twin framework. While, on-site monitoring data directly reflect the actual operating state of the dam and therefore constitute high-fidelity (HF) observations. Nevertheless, such data are often constrained by limited sample size, measurement noise, and incomplete coverage of extreme operating conditions. Moreover, purely data-driven models trained only on monitoring data may generate predictions that lack physical plausibility when extrapolated beyond the historical data domain.

To simultaneously ensure physical consistency and engineering accuracy, this study develops a physics-guided multi-fidelity Bayesian neural network (PGMF-BNN) framework, in which low-fidelity data are first used to establish a fundamental mechanical response skeleton, and high-fidelity data are subsequently introduced to correct the discrepancy between simulation and reality. On this basis, Bayesian inference is incorporated to generate probabilistic predictions for adaptive monitoring benchmarks.

The low-fidelity dataset D_{LF} is defined as:

$$D_{LF} = \left\{ \left(\mathbf{x}_{phy}^{(i)}, \mathbf{y}_{LF}^{(i)} \right) \right\}_{i=1}^{N_{LF}} \tag{3}$$

where i is the index of the *LF* sample, $i = 1, 2, \cdots, N_{LF}$;N_{LF} is the total number of *LF* samples; $\mathbf{x}_{phy}^{(i)}$ is the physical input vector of the $i-$th *LF* sample, used to describe the mechanical state dominated by water pressure; $\mathbf{y}_{LF}^{(i)} \in \mathbb{R}^N$ represents the displacement response vector corresponding to the $i-$th *LF* sample, which is calculated by the FEM, and its dimension n matches that of the multi-point output. The purpose of this dataset is to provide an initial mechanically informed representation for the digital twin model, rather than to serve directly as the final monitoring truth.

The high-fidelity dataset D_{HF} is defined as:

$$D_{HF} = \left\{ \left(\mathbf{x}_{phy}^{(j)}, \mathbf{x}_{env}^{(j)}, \mathbf{y}_{HF}^{(j)} \right) \right\}_{j=1}^{N_{HF}} \tag{4}$$

where j is the index of the *HF* sample, $j = 1, 2, \cdots, N_{HF}$;N_{HF} is the total number of *HF* samples; $\mathbf{x}_{phy}^{(j)}$ is the physical input vector of the $j-$th *HF* sample; $\mathbf{x}_{env}^{(j)}$ is the environmental/time-effect input vector of the $j-$th *HF* sample, which characterizes temperature, aging, and other sources of deviation; $\mathbf{y}_{HF}^{(j)} \in \mathbb{R}^N$ represents the measured displacement vector of the $j-$th *HF* sample, sourced from on-site monitoring data. High-fidelity data serve to correct deviation and update the true state.

Accordingly, the overall digital twin prediction model is formulated as an additive mapping consisting of a physics-based main-response branch $\mathbf{f}_{phy}$ and a residual-correction branch $\mathbf{f}_{res}$, the expression is exhibited as

$$\widehat{\mathbf{y}}(t) = \mathbf{f}_{phy}\left(\mathbf{x}_{phy}(t); \theta_p\right) + \mathbf{f}_{res}(\mathbf{x}_{env}(t); \theta_e) + \mathbf{b} \tag{5}$$

where t is the time index, consistent with the monitoring time series; $\widehat{\mathbf{y}}(t)$ represents the model-predicted displacement vector at time t; $\mathbf{x}_{phy}(t)$, $\mathbf{x}_{env}(t)$ have the same definitions as previously described. The function $\mathbf{f}_{phy}\left(\cdot; \theta_p\right)$ denotes the nonlinear mapping implemented by the physical branch, whose objective is to learn the fundamental deformation response skeleton governed by structural mechanics. The function $\mathbf{f}_{res}(\cdot; \theta_e)$ denotes the nonlinear mapping implemented by the residual branch, which is designed to compensate for the systematic discrepancy between the finite element-based physical response and the actual monitored response. $\mathbf{b} \in \mathbb{R}^N$ is a learnable bias vector introduced to account for the overall baseline deviation.

Eq. (5) defines an additive digital twin mapping that integrates a physics-based main-response component with a data-driven residual-correction component. The formulation is built upon a two-stage knowledge transfer process, in which low-fidelity data are used to establish a mechanically meaningful response skeleton at first, and high-fidelity monitoring data are subsequently introduced to correct the virtual–physical discrepancy. As a result, the proposed framework enables physically interpretable predictions while maintaining adaptive updating capability under evolving operating conditions.

3.2.2. Physics-guided two-stage transfer learning for virtual–physical alignment

To address the discrepancy between simulation-based representations and real structural responses, a physics-guided two-stage transfer learning strategy is developed for virtual–physical alignment. Based on the additive digital twin formulation in Section 3.2, the proposed approach separates the learning process into two sequential stages with clearly differentiated yet complementary roles. In Stage I, low-fidelity simulation data are used to establish a mechanically consistent response skeleton, providing a physics-informed initialization of the model. In Stage II, high-fidelity monitoring data are introduced to correct the virtual–physical discrepancy through residual learning, enabling adaptive refinement toward real operating conditions. This staged learning mechanism ensures that physical consistency is preserved while allowing data-driven correction under evolving conditions.

The detailed formulations of the two stages are presented in Sections 3.2.2.1 and 3.2.2.2, respectively.

3.2.2.1. Stage I: Physics-guided pre-training. The objective of Stage I is to establish a mechanically consistent response representation. Specifically, low-fidelity simulation data are employed to learn a fundamental response skeleton in the virtual space that conforms to the governing laws of structural mechanics[34,35]. To prevent the residual branch from learning non-transferable bias from low-fidelity data, only the physical branch is optimized in this stage. The residual branch associated with environmental variables is kept frozen, and only the parameters of the physical branch θ_p and the bias term b are updated. Under this setting, the model output can be interpreted as the baseline theoretical response of the digital twin under dominant loading conditions:

$$\widehat{\boldsymbol{y}}_{LF}^{(i)} = \boldsymbol{f}_{phy}\left(\boldsymbol{x}_{phy}^{(i)}; \boldsymbol{\theta}_p\right) + \boldsymbol{b} \tag{6}$$

In Stage I, the mean squared error (*MSE*) was used as the training loss to optimize the physical branch and the bias term, and the corresponding loss function is defined as:

$$L_{P1}\left(\theta_p, \mathbf{b}\right) = \frac{1}{N_{LF}N} \sum_{i=1}^{N_{LF}} \left\| \mathbf{y}_{LF}^{(i)} - \widehat{\mathbf{y}}_{LF}^{(i)} \right\|_2^2 \tag{7}$$

where N denotes the number of monitoring points.

3.2.2.2. Stage II: Residual correction and virtual–physical alignment. In Stage II, high-fidelity monitoring data are introduced to refine the model by correcting the discrepancy between the virtual representation and the actual structural response. Unlike Stage I, both the physical branch and the residual branch participate in the training process. The residual branch is primarily responsible for learning the environmental and time-dependent effects, while the physical branch is retained to preserve the established mechanical response skeleton.

To achieve effective correction without degrading the learned physical representation, a differential learning rate strategy is adopted:

$$\eta_{phy} \ll \eta_{res} \tag{8}$$

where the smaller learning rate η_{phy} is used solely to update the parameters θ_p of the physical branch, while the larger learning rate η_{res} is used to update the parameters θ_e of the residual branch; $\mathbf{b}$ is updated alongside η_{res}. This strategy allows the residual branch to take the dominant role in correcting the virtual–physical mismatch, while the physical branch is adjusted only incrementally to maintain mechanical consistency.

In Stage II, the mean squared error (*MSE*) was also used as the data-fitting loss for virtual–physical alignment, and the corresponding loss function L_{P2} is defined as:

$$L_{P2}(\theta_p, \theta_e, \mathbf{b}) = \frac{1}{N_{HF}N} \sum_{j=1}^{N_{HF}} \left\| \mathbf{y}_{HF}^{(j)} - \widehat{\mathbf{y}}^{(j)} \right\|_2^2 \tag{9}$$

To further avoid non-physical predictions under complex or extreme operating conditions, a parameter regularization constraint L is introduced for the physical branch:

$$L = L_{P2} + \lambda \left\| \theta_p - \theta_p^{(P1)} \right\|_2^2 \tag{10}$$

where λ represents the weighting coefficient, which is selected based on the validation set; $\theta_p^{(P1)}$ denotes the converged physical branch parameters obtained from Stage I. This constraint prevents the physical branch from deviating excessively from the learned mechanical skeleton, thereby reducing the risk of non-physical predictions.

Through the above two-stage transfer learning process, the model achieves a balance between physical consistency and data-driven adaptability, enabling accurate and physically interpretable prediction of dam deformation under evolving operating conditions.

3.2.3. Mimo-based spatial collaborative modeling

To characterize the spatial coupling among deformation responses at multiple monitoring points of a concrete arch dam, a multi-input multi-output (MIMO)[25,36] collaborative prediction architecture is further introduced. The input layer incorporates multi-source variables, including hydraulic loading, thermal conditions, time-effect factors, and material parameters of different dam sections, while the output layer corresponds to the radial displacement responses at multiple monitoring points. In contrast to conventional single-point regression, this architecture is designed to capture the overall structural response pattern of the dam through joint modeling of spatially distributed observations.

Based on the additive digital twin formulation in Eq. (5), the input variables are partitioned into two categories according to their physical semantics and are fed into the physical branch and the residual branch, respectively.

(1) Physical branch input

The input of the physical branch, denoted by $\mathbf{x}_{phy}(t)$ consists of the dominant loading and material-related variables:

$$\mathbf{x}_{phy}(t) = [H(t), E_1, E_2, E_3, E_4]^{\mathrm{T}} \in \mathbb{R}^5 \tag{11}$$

where $H(t)$ represents the upstream water level, and $E_1 \sim E_4$ denote the elastic moduli of the four dam sections. These parameters are introduced to reflect the influence of material heterogeneity on the global deformation mode of the dam. Other structural and foundation parameters are treated as fixed according to the design and numerical simulation settings.

(2) Residual branch input

The input of the residual branch, denoted by $\mathbf{x}_{env}(t)$, is used to characterize environmental and time-dependent effects:

$$\mathbf{x}_{env}(t) = \left[T_{cur}(t), T_{lag30}(t), \ln(t+1)\right]^{\mathrm{T}} \in \mathbb{R}^3 \tag{12}$$

where $T_{cur}(t)$ represents the current temperature, $T_{lag30}(t)$ denotes the 30-day moving-average temperature used to characterize thermal inertia, and $\ln t + 1$ is the logarithmic time factor used to represent long-term convergent evolution. Here, t denotes the elapsed monitoring time interval from the initial monitoring date, with $t = 0$ corresponding to October 25, 2013.

Accordingly, the total input at time t can be written as:

$$\mathbf{x}(t) = \left[\mathbf{x}_{phy}(t), \mathbf{x}_{env}(t)\right] \tag{13}$$

For the output side, the radial displacements at the ten monitoring points are organized into a unified displacement vector:

$$\mathbf{y}(t) = \left[y_1(t), y_2(t), \ldots, y_{10}(t)\right]^{\mathrm{T}} \in \mathrm{R}^N \tag{14}$$

where $y_k(t)(k = 1, 2, \cdots, N)$ represents the observed radial displacement at the $k-$th monitoring point at time t. In this study, the MIMO output dimension was set as $N = 10$, including six plumb line monitoring points and four inverted plumb line monitoring points. Therefore, the proposed model learns a ten-point deformation response vector at each time step.

Unlike conventional approaches that model each monitoring point independently, the proposed MIMO architecture employs a unified output layer to perform joint regression over all monitoring points. This design allows the network to learn shared latent representations of the structural deformation field, thereby capturing both the global deformation pattern induced by hydraulic loading and the spatially varying responses associated with thermal and time-dependent effects. As a result, the model is able to describe the coordinated evolution of the overall structural state, rather than generating a set of isolated local predictions.

3.3. Probabilistic prediction and uncertainty quantification

3.3.1. Bayesian neural network for deformation prediction

In practical dam operation, deformation prediction is affected not only by the uncertainty of model parameters and network structure, but also by measurement noise and environmental disturbances. Consequently, deterministic models that provide only point estimates cannot fully satisfy the requirements of digital twin-based safety monitoring, where both predictive accuracy and reliability assessment are essential [37–39]. To address this issue, the proposed framework adopts a Bayesian neural network (BNN)[40,41] to extend deterministic prediction to probabilistic inference. By treating network weights as random variables, the BNN provides a rigorous mathematical framework for capturing both epistemic uncertainties associated with model parameters and aleatoric uncertainty arising from observation noise, thereby enabling the construction of reliable prediction intervals for dam deformation monitoring.

In a deterministic neural network, the model parameters $\theta = (\theta_p, \theta_e, \mathbf{b})$ are treated as fixed values, and the output is a point prediction $\widehat{\mathbf{y}}(t)$.

By contrast, in the Bayesian formulation, the network weights are modeled as random variables $\mathbf{w}$, and the prediction is conditioned on their posterior distribution. Given the input $\mathbf{x}(t) = [\mathbf{x}_{phy}(t), \mathbf{x}_{env}(t)]$, the model output can be written as

$$\hat{\mathbf{y}}(t) = f(\mathbf{x}(t); \mathbf{w}) \tag{15}$$

where $f(\cdot)$ denotes the dual-branch additive network defined in Eq. (5).

Assuming that the observation noise follows a Gaussian distribution, the multi-output likelihood function can be expressed as

$$p(\mathbf{y}(t), \mathbf{x}(t), \mathbf{w}) = N(\mathbf{y}(t); f(\mathbf{x}(t); \mathbf{w}), \sigma_n^2 \mathbf{I}) \tag{16}$$

where $\mathbf{y}(t) \in \mathrm{R}^N$ denotes the observed displacement vector, σ_n^2 is the noise variance; $\mathbf{I}$ is the identity matrix of size $N \times N$.

The objective of Bayesian inference is to estimate the posterior distribution of the network weights conditioned on the observed data D, which can be written as:

$$p(\mathbf{w}|D) \propto p(D|\mathbf{w}) p(\mathbf{w}) \tag{17}$$

However, because the exact posterior distribution $p(\mathbf{w}|D)$ in Bayesian neural networks is generally intractable, direct Bayesian inference is not feasible in practice. To address this issue, a tractable variational distribution $q_\phi(\mathbf{w})$ is introduced to approximate the true posterior, and the approximation is obtained by minimizing the Kullback–Leibler (KL)[42] divergence between the two distributions.

$$\mathrm{KL}\big(q_\phi(\mathbf{w}) \| p(\mathbf{w}|D)\big) \tag{18}$$

where KL denotes the Kullback-Leibler divergence, which measures the difference between two probability distributions. The KL divergence term plays a regularization role, which will punish the situation that the variational distribution deviates too much from the prior distribution, thus effectively preventing the model from overfitting and capturing the uncertainty.

3.3.2. Uncertainty quantification via MC Dropout

To achieve a practical balance between computational efficiency and probabilistic inference in engineering applications, MC Dropout[43] is adopted to approximate the Bayesian prediction process described above. The essential idea of MC Dropout is to retain dropout activation during inference, such that each stochastic forward pass corresponds to sampling a set of effective network weights from the approximate posterior distribution $q_\phi(\mathbf{w})$. In this way, the predictive distribution can be efficiently estimated via Monte Carlo sampling without substantial modification of the network architecture or excessive computational overhead, thereby making MC Dropout particularly suitable for large-scale digital twin applications involving multi-point prediction and online updating.

For a given input $\mathbf{x}(t)$ at time t, T stochastic forward passes yield a set of predictive samples:

$$\hat{\mathbf{y}}^{(s)}(t) = f(\mathbf{x}(t); \mathbf{w}^{(s)}), s = 1, 2, \cdots, T \tag{19}$$

$$\hat{\mathbf{y}}^{(s)}(t) \in \mathrm{R}^N$$

The predicted mean for multi-output vectors $\hat{\mu}(t)$ is:

$$\hat{\mu}(t) = \frac{1}{T} \sum_{s=1}^{T} \hat{\mathbf{y}}^{(s)}(t) \in \mathrm{R}^{D_{out}} \tag{20}$$

For the multi-output prediction case considered in this study, epistemic uncertainty is characterized by the sample covariance of the stochastic predictions as:

$$\sum{}_{epi}(t) = \frac{1}{T} \sum_{s=1}^{T} \big(\hat{\mathbf{y}}^{(s)}(t) - \hat{\mu}(t)\big)\big(\hat{\mathbf{y}}^{(s)}(t) - \hat{\mu}(t)\big)^{\mathrm{T}} \tag{21}$$

$$\sum{}_{epi}(t) \in \mathrm{R}^{N \times N}$$

Assuming homoscedastic observation noise, the aleatoric uncertainty is represented as

$$\sum{}_{ale}(t) = \sigma_n^2 \mathbf{I} \tag{22}$$

The total uncertainty $\sum_{tot}(t)$ is:

$$\sum{}_{tot}(t) = \sum{}_{epi}(t) + \sum{}_{ale}(t) \tag{23}$$

In the case of point-by-point output, the confidence interval for the k−th monitoring point $CI_{95\%,k}(t)$ is represented as

$$CI_{95\%,k}(t) : \hat{\mu}_k(t) \pm 1.96 \sqrt{\sum{}_{tot,kk}(t)} \tag{24}$$

where $\sum_{tot,kk}(t)$ represents the diagonal elements of $\sum_{tot}(t)$. The above formulation provides dynamic prediction intervals for each monitoring point and thus enables the construction of state-dependent probabilistic monitoring benchmarks.

3.4. Implementation and deployment of the digital twin prediction framework

To illustrate the implementation procedure of the proposed framework, the technical route for digital twin-based deformation prediction of concrete arch dams is summarized in Fig. 3.

The procedure consists of seven interconnected stages. First, low-fidelity simulation data and high-fidelity monitoring data are constructed to form a multi-source heterogeneous dataset. Second, a dual-branch additive mapping model is established, in which the physical branch captures the dominant mechanical response and the residual branch corrects the virtual–physical discrepancy. Third, the physical branch is pre-trained using low-fidelity data to establish a mechanically meaningful response skeleton. Fourth, high-fidelity monitoring data are introduced for transfer learning, where the residual branch is activated and the virtual model is aligned with actual operating conditions through differential learning rates and physical regularization. Fifth, a MIMO-based collaborative prediction mechanism is constructed to capture the spatial coupling among multiple monitoring points. Sixth, Bayesian inference with MC Dropout is incorporated to achieve probabilistic prediction and uncertainty quantification. Finally, dynamic monitoring benchmarks are generated from the predictive mean and uncertainty intervals to support online trend inference, anomaly detection, and risk early warning. Overall, the proposed framework provides an integrated technical route for physically interpretable, adaptively updated, and uncertainty-aware digital twin prediction of arch dams.

To further improve the reproducibility of the proposed framework, the implementation details and key hyperparameters used in model training and probabilistic inference are summarized in Table 1. The hyperparameters include the network architecture, optimizer, loss function, learning rates in the two-stage training process, dropout setting, batch size, and the number of MC Dropout samples. These settings were adopted consistently in the following case study and model comparison experiments.

Full-batch training was adopted in both stages because the sample size of the low-fidelity simulation data and high-fidelity monitoring data was relatively limited. In Stage I, all low-fidelity samples were used in each epoch to learn the mechanically consistent response skeleton. In Stage II, all high-fidelity training samples were used in each epoch for virtual–physical alignment and residual correction.

4. Project examples

The proposed framework was validated using a real concrete arch

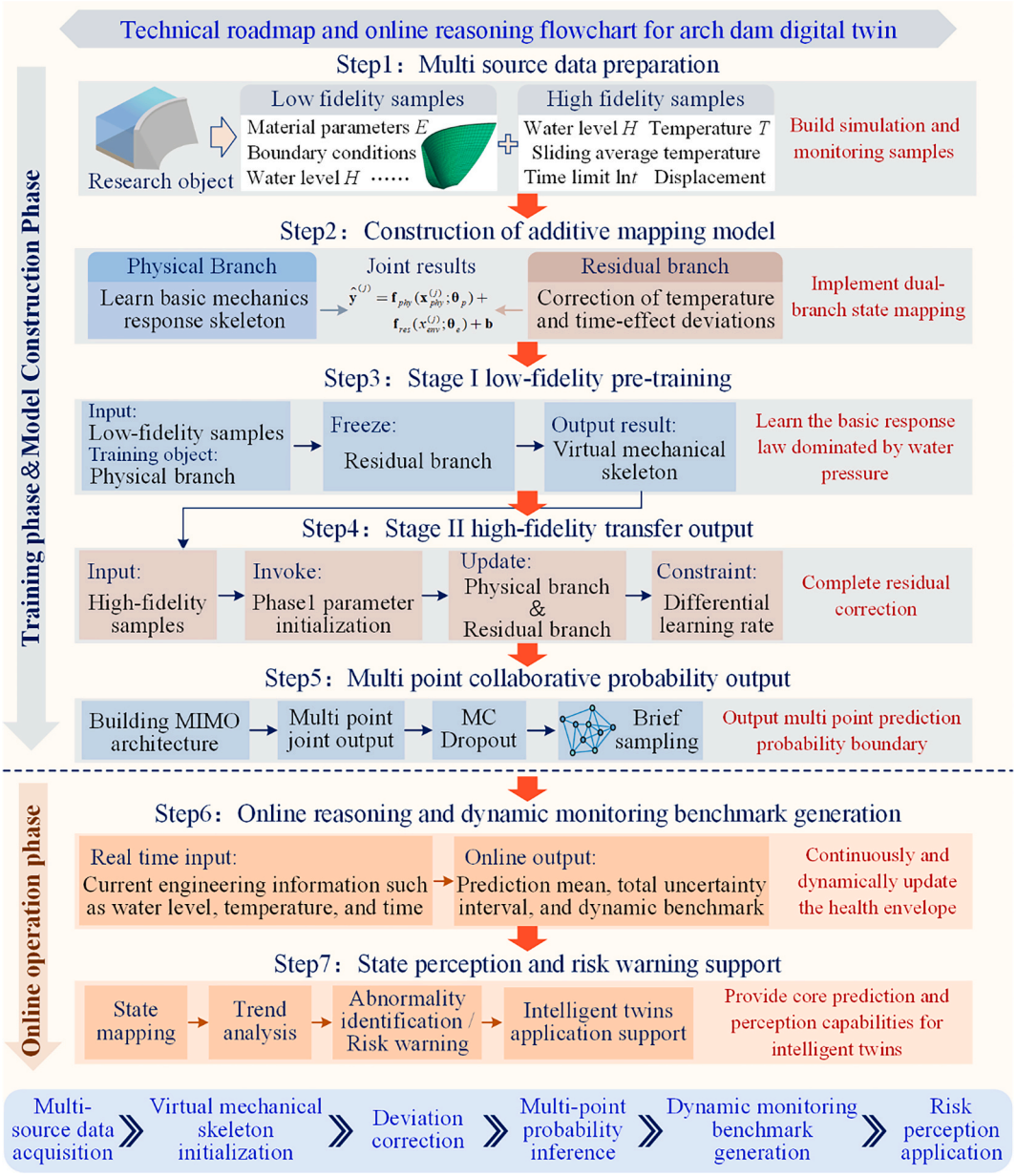


Fig. 3. Flowchart of the proposed prediction framework for digital twin monitoring.

dam project, namely the Shankouyan Reservoir in Jiangxi Province, China. The reservoir is located in Shankouyan Village, Pingxiang City, in the upper reaches of the Yuan River, which controls a catchment area of 230 km^2, with a total storage capacity of $1.05 \times 10^8 m^3$. The dam adopts a roller-compacted concrete (RCC) double-curvature arch dam scheme, with a maximum dam height of 99.1 m. Construction started in November 2007, and the reservoir began impoundment in June 2012. The installed hydropower capacity is 13.26 MW, with an average annual power generation of 3.36×10^7 kWh and a daily water supply capacity of 2.0×10^5 m^3.

4.1. Engineering background, FEM modeling and data correlation analysis

To support long-term safety monitoring, a comprehensive instrumentation system has been deployed within the dam, including plumb lines, inverted plumb lines, and temperature sensors. The spillway section is the key area for displacement monitoring, where two plumb-line monitoring points (PLA1 and PLA2) and two inverted plumb-line monitoring points (IPA1 and IPA2) have been installed, as shown in Fig. 4 (a). These measurements provide the high-fidelity monitoring data used in this study for model calibration and validation.

To generate the low-fidelity simulation data required for physics-guided pre-training, a three-dimensional finite element model of the dam–foundation system was established, as shown in Fig. 4 (b)~(c). The coordinate system of the model was defined such that the radial direction corresponds to the *x*-axis, with the downstream direction taken as positive; the tangential direction corresponds to the *y*-axis, with the left bank taken as positive and the vertical direction corresponds to the *z*-axis, with upward taken as positive. To reduce boundary effects, the computational domain of the foundation extended to a depth of twice the dam height, and to distances of 1.0 and 1.5 times the dam height in the upstream and downstream directions, respectively. The finite element model was discretized primarily using hexahedral and tetrahedral elements, with a total of 151,200 mesh cells. Since the dam utilized in this research is an RCC arch dam constructed by layered compaction, the dam body exhibits evident zoning characteristics after construction. To better represent the material heterogeneity of the

Table 1
Hyperparameter settings of the proposed PGMF-BNN model.

Category	Hyperparameter	Value/setting
General setting	Programming framework	PyTorch
General setting	Optimizer	Adam
General setting	Loss function	Mean squared error
General setting	Random seed	42
Network architecture	Physical branch input	$H(t), E_1, E_2, E_3, E_4$
Network architecture	Residual branch input	$T_{cur}(t), T_{lag30}(t), \ln(t+1)$
Network architecture	Output dimension	10 monitoring points
Network architecture	Physical branch structure	5–128–64–10
Network architecture	Physical branch activation	Tanh
Network architecture	Residual branch structure	3–64–10
Network architecture	Residual branch activation	ReLU
Bayesian approximation	Dropout rate	0.05
Bayesian approximation	Number of MC Dropout samples	100
Stage I: physics-guided pre-training	Optimized parameters	Physical branch and bias term
Stage I: physics-guided pre-training	Learning rate	0.005
Stage I: physics-guided pre-training	Epochs	2000
Stage I: physics-guided pre-training	Batch size	Full-batch training
Stage II: residual correction and fine-tuning	Optimized parameters	Physical branch, residual branch, and bias term
Stage II: residual correction and fine-tuning	Learning rate of physical branch	0.0005
Stage II: residual correction and fine-tuning	Learning rate of residual branch	0.002
Stage II: residual correction and fine-tuning	Learning rate of bias term	0.002
Stage II: residual correction and fine-tuning	Epochs	3000
Stage II: residual correction and fine-tuning	Batch size	Full-batch training

dam–foundation system, the foundation was divided into two zones according to the geological survey data of the site, while the dam body was partitioned into four zones based on construction and material records. The subsequent forward simulations were conducted on this basis to provide low-fidelity displacement samples for the proposed multi-fidelity digital twin framework.

To characterize the effects of key environmental variables on dam deformation, the temporal evolution and statistical distributions of upstream water level, average temperature, and radial displacement at representative monitoring points were analyzed, as shown in Fig. 5.

As exhibited in Fig. 5(a), the upstream water level fluctuated mainly within the range of approximately 205–245 m during the study period. A relatively low-water stage occurred in early 2014, followed by a rapid increase, after which the reservoir generally operated within a stable interval. The cumulative frequency distribution further indicates that the middle water-level range (231.1–242.5 m) accounts for nearly 80% of the observations, whereas both low- and high-water conditions each account for only about 10%. In Fig. 5(b), it is presented that the average temperature exhibits a pronounced annual periodic pattern, with higher values in and lower values in winter, indicating a stable seasonal thermal regime throughout the observation period. No evident long-term drift or abnormal fluctuation is observed. The cumulative distribution shows that the middle temperature range (8.9–29.3 °C) accounts for approximately 79.3% of the samples, while low- and high-temperature conditions represent relatively small proportions. This distribution indicates that the thermal boundary conditions during the study period were also concentrated mainly within the normal seasonal range.

The radial displacement responses of the six key monitoring points, shown in Fig. 5(c), display clear periodic fluctuations over time. The displacement curves exhibit similar temporal patterns across different points, although their amplitudes differ noticeably. In particular, the monitoring points in the PLA zone show larger displacement amplitudes than those in the PLB and PLC zones, suggesting that the deformation response is spatially heterogeneous. Nevertheless, the consistency in temporal variation among the monitoring points indicates that the dam deformation field retains a strong degree of overall coordination.

The Pearson correlation analysis in Fig. 6 further reveals the dominant relationships among hydraulic, thermal, and deformation variables. Upstream water level shows a generally positive correlation with radial displacement, with correlation coefficients ranging from 0.30 to 0.74, confirming that hydraulic loading is an important driver of structural deformation, although its influence varies across locations. In contrast, average temperature exhibits an overall negative correlation with the displacement responses, with coefficients approximately ranging from − 0.01 to − 0.56, reflecting the cooling–warming effect associated with temperature-induced deformation. More importantly, the deformation responses among different monitoring points are highly correlated, with most correlation coefficients exceeding 0.75 and several approaching 0.98, which demonstrates strong spatial coherence and coordinated structural behavior. These observations provide direct evidence for the necessity of introducing a multi-point collaborative prediction framework in subsequent modeling.

4.2. Results of physics-guided multi-fidelity MIMO framework of deformation prediction

4.2.1. Physics-guided two-stage transfer learning

The proposed PGMF-BNN was trained in a MIMO manner with ten monitoring points as outputs. For quantitative performance evaluation, six representative plumb line monitoring points located at the dam crest and dam waist were selected, namely PLA2, PLB2, PLC2, PLA1, PLB1, and PLC1. The inverted plumb line points, including IPA2, IPA1, IPB1, and IPC1, were retained in the MIMO output but were not used for regression-based quantitative evaluation because their displacement amplitudes are comparatively small and more susceptible to measurement noise.

To accommodate the two-stage training strategy of PGMF-BNN, the model inputs were divided into two groups according to the physical causes of dam deformation, namely physics-related drivers and deviation-correction variables. The first group was used primarily in Stage I for physical pre-training, with the objective of enabling the model to learn the fundamental mechanical response governed by structural mechanics. Since the upstream water level H is the dominant external load, its admissible range was first determined from historical monitoring records $\left[H_{\min}^{history}, H_{\max}^{history}\right]$. To improve the model's ability to cover operating conditions beyond the historical range, additional boundary values, design flood level, crest elevation, and historical extreme water levels were incorporated. Accordingly, the final water-level interval $[H_{\min}, H_{\max}]$ was established. In addition, the elastic moduli $E_1 \sim E_4$ of the four material zones in the dam body were selected as the key physical parameters in Stage I, and their variation ranges were defined according to design values and related engineering data. These parameters determine the deformation resistance of the dam in the finite element simulations and therefore provide essential physical information for learning the nonlinear mapping between stiffness and displacement during pre-training. The corresponding ranges of the key variables are listed in Table 2. In this study, the Latin hypercube sampling (LHS) method was used to generate 1,000 low-fidelity samples for

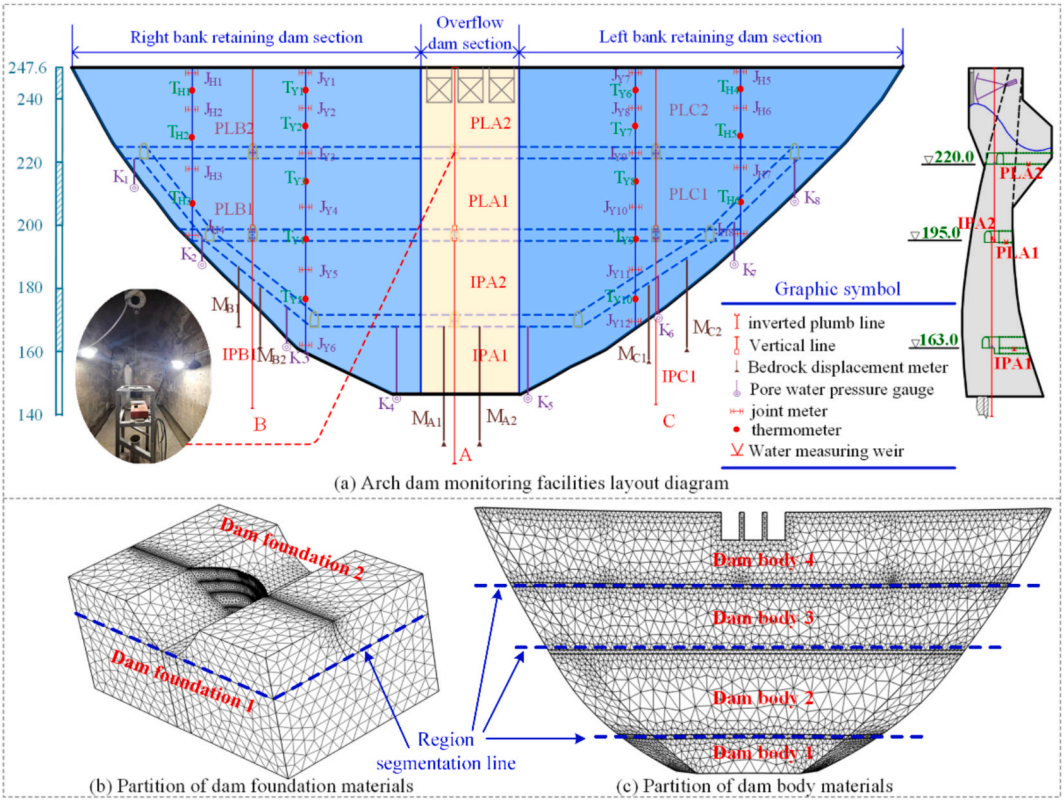


Fig. 4. Layout of monitoring points and 3D finite element model of Shankouyan RCC arch dam.

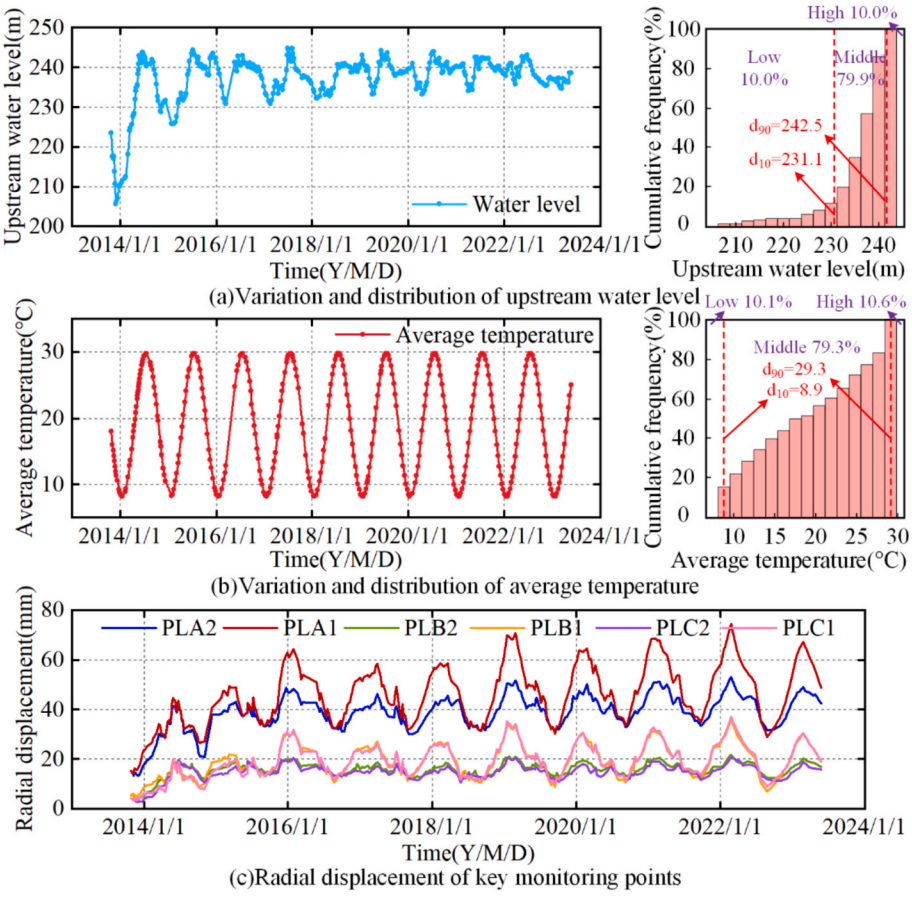


Fig. 5. Spatio-temporal correlation analysis of environmental factors and displacement.

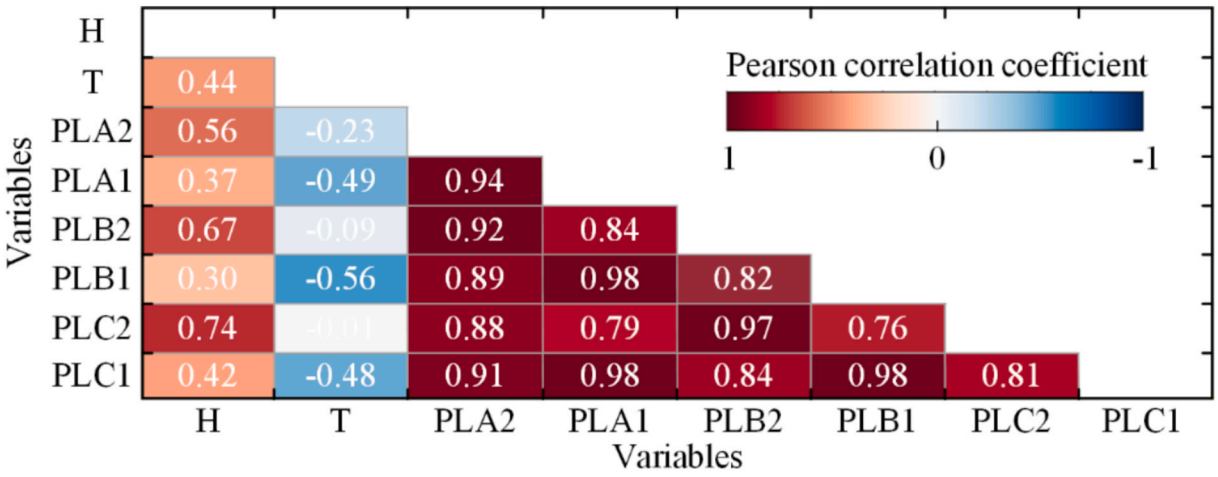


Fig. 6. Correlation analysis among environmental factors and displacement.

Table 2
Range of values for key variables required by the model during the pre-training phase.

Key variables	H(m)	E_1(GPa)	E_2(GPa)	E_3(GPa)	E_4(GPa)
Maximum value of interval	247	32.4	30	27	24
Design value	/	27	25	22.5	20
Minimum value of interval	200	21.6	20	18	16

Stage I training. Other parameters with relatively minor influence on dam deformation, such as the deformation modulus of the foundation, the Poisson's ratios of the dam and foundation materials, and the unit weight, were treated as fixed values in this stage. The specific values are shown in Table 3.

To visually examine the ability of the finite element model to capture the macroscopic mechanical response of the dam, representative simulation results under typical operating conditions were selected for contour analysis. As shown in Fig. 7, the radial displacement contours of the upstream dam surface are arranged in a 3 × 3 layout. From top to bottom, the three rows correspond to typical low-, medium-, and high-water conditions, i.e., 205 m, 225 m, and 245 m, respectively. From left to right, the three columns correspond to three representative combinations of elastic moduli, reflecting relatively soft, normal, and stiff material states. To facilitate visual comparison of global deformation patterns under different conditions, each contour plot adopts the same viewing perspective, while its color scale is adjusted individually to ensure adequate sensitivity to displacement variations under both mild and unfavorable conditions.

Table 3
Fixed finite-element parameters used for low-fidelity data generation.

Parameter	Value	Unit
Deformation modulus of dam foundation 1	12.0	GPa
Deformation modulus of dam foundation 2	16.0	GPa
Poisson's ratio of dam body 1	0.25	–
Poisson's ratio of dam body 2	0.22	–
Poisson's ratio of dam body 3	0.19	–
Poisson's ratio of dam body 4	0.16	–
Poisson's ratio of dam foundation 1	0.35	–
Poisson's ratio of dam foundation 2	0.25	–
Unit weight of dam body 1	25.5	kN/m^3
Unit weight of dam body 2	24.5	kN/m^3
Unit weight of dam body 3	23.5	kN/m^3
Unit weight of dam body 4	22.5	kN/m^3
Unit weight of dam foundation 1	26.0	kN/m^3
Unit weight of dam foundation 2	22.0	kN/m^3

As shown in Fig. 7, the radial displacement of the arch dam increases markedly with increasing water level and decreases with increasing elastic modulus, which is consistent with the expected mechanical behavior of the structure. The combination of high water level and relatively low elastic modulus produces the maximum overall displacement response, corresponding to the most unfavorable deformation state among the selected cases. By contrast, the combination of low water level and high elastic modulus yields the minimum displacement response. These results indicate that the low-fidelity finite element samples can effectively characterize the deformation evolution of the dam under different hydraulic and material conditions, thereby providing a physically meaningful response skeleton for subsequent multi-fidelity learning.

The second group of variables was introduced in Stage II to characterize environmental disturbances and long-term time-dependent effects that cannot be explicitly represented by the physics-based branch. In addition to the current ambient temperature T_{cur}, which reflects the instantaneous thermal boundary condition, the 30-day moving-average temperature T_{lag30} was further introduced to account for the pronounced thermal inertia of the concrete dam body, thereby describing the lagged and cumulative influence of ambient temperature variation on the internal thermal state of the structure. To justify the selection of the moving-average temperature window, a sensitivity analysis was conducted using candidate windows of 15, 30, 45, 60, and 90 days. The validation-set performance under different window lengths is summarized in Table 4.

As shown in Table 4, the 30-day moving-average temperature achieved the best validation performance, with the lowest average RMSE of 1.3421 mm and the highest average R^2 of 0.9281. Although the 15-day window also produced comparable results, longer windows of 45–90 days led to a noticeable decrease in predictive accuracy, indicating that excessive smoothing may weaken the representation of short- to medium-term thermal-lag effects. Therefore, T_{lag30} was adopted in the residual branch to characterize the lagged and cumulative influence of ambient temperature variation.

To represent irreversible long-term effects, such as creep and long-term foundation deformation, the logarithmic time factor ln$t$ was also incorporated as an input variable. Through these variables, the residual branch is able to compensate for the discrepancy between the physics-based response and the measured structural behavior induced by environmental and time-dependent factors.

Based on the available monitoring records, a total of 349 daily samples collected between October 25, 2013 and May 25, 2023 were selected to construct the high-fidelity dataset. The samples were divided into the training, validation, and testing sets in a ratio of 8:1:1, following

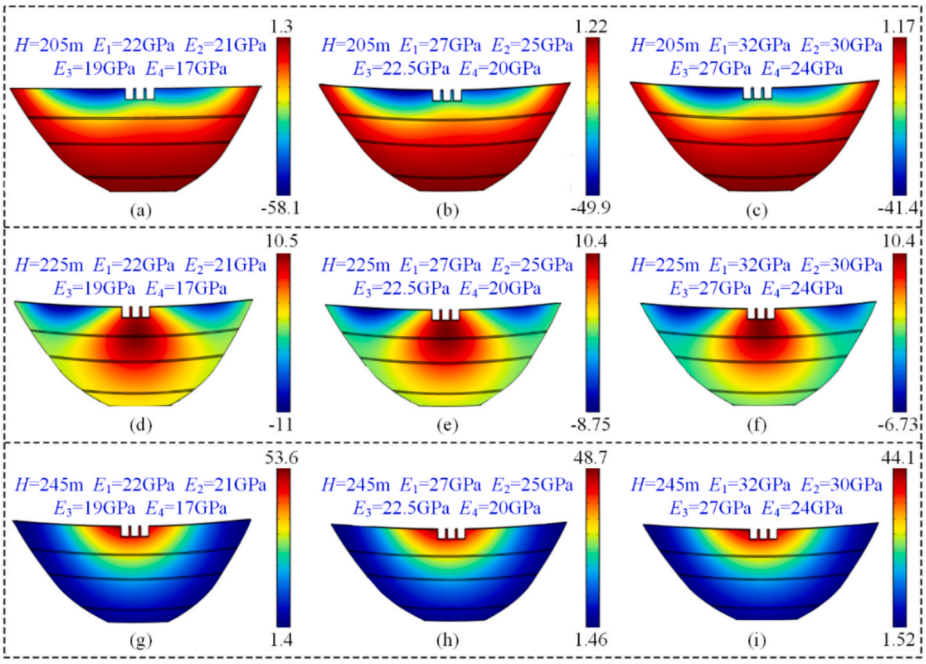


Fig. 7. Radial displacement contour of upstream surface of arch dam.

Table 4
Sensitivity analysis of moving-average temperature window length.

Moving-average window	Average RMSE(mm)	Average R^2	Time(s)
15 days	1.3466	0.9251	16.11
30 days	**1.3421**	**0.9281**	17.18
45 days	1.9267	0.8766	15.65
60 days	1.9133	0.8806	**14.98**
90 days	1.6604	0.8778	24.44

chronological order to avoid information leakage. On this basis, further analysis was conducted from a spatial perspective to assess whether the MIMO architecture can effectively capture the correlation structure and coordinated deformation behavior among multiple monitoring points.

4.2.2. *Mimo-based modeling results of deformation prediction*

Here, a MIMO-based prediction model was constructed to capture the spatial topological relationship between multiple monitoring points of arch dam. As shown in Fig. 8(a) and Fig. 8(b), the predicted values of the six key monitoring points at the dam crest and dam waist agree well with the corresponding measurements in both the training and validation sets. The coefficients of determination R^2 are generally higher than 0.90, except for PLC2 and PLC1 in the validation set, where the values are 0.827 and 0.876, respectively. These results indicate that the proposed model achieves satisfactory generalization performance without evident underfitting or overfitting. The regression results on the testing set, shown in Fig. 9, further confirm the high predictive accuracy of the model and demonstrate that the proposed framework can effectively capture the dominant deformation modes of the dam. For the deep bedrock monitoring points (IP points), the actual deformation amplitudes are relatively small, and the signals are close to the measurement noise level. Although the model outputs for these points remain generally stable, the corresponding evaluation metrics are more sensitive to minor perturbations. Therefore, regression-based quantitative analysis was not performed for these points.

To further assess the ability of the model to represent spatial structural relationships, the measured and reconstructed spatial correlation matrices were compared, as shown in Fig. 10. The correlation matrix predicted by PGMF-BNN reproduces the overall distribution pattern of the measured matrix with high fidelity. In particular, the locations of high-correlation and low-correlation regions are well preserved, indicating that the model has successfully captured the intrinsic spatial association among different monitoring points. Correlation result suggests that the proposed framework is capable not only of accurate multi-point prediction, but also of learning the coordinated deformation behavior governed by the overall topological structure of the arch dam.

In addition, the computational cost was evaluated to examine the applicability of the proposed framework for digital twin-based monitoring. The timing test was conducted using the same implementation as that adopted in the case study, including the two-stage training process and MC Dropout-based probabilistic inference. The corresponding hardware configuration and computational cost are listed in Table 5.

As shown in Table 5, the Stage II fine-tuning process required 6.364 s, while the MC Dropout-based probabilistic inference with 100 stochastic forward passes required only 25.666 ms for the testing set, corresponding to an average inference time of 0.733 ms per monitoring sample. These results indicate that the proposed framework is computationally feasible for daily-scale digital twin-based dam deformation monitoring.

Although the present case study implements the proposed framework in an offline training–testing manner, its two-stage learning mechanism provides a feasible basis for online updating in digital twin deployment. After Stage I low-fidelity pre-training and Stage II high-fidelity fine-tuning, newly collected monitoring data can be used for probabilistic multi-point deformation prediction through MC Dropout, with the predictive mean and uncertainty interval serving as dynamic monitoring benchmarks. In practical deployment, if newly accumulated monitoring data show persistent prediction residuals or reduced interval coverage within a recent time window, Stage II fine-tuning can be reactivated. Consistent with the differential learning-rate strategy used in this study, the residual branch and bias term can be preferentially updated to correct new virtual–physical discrepancies, while the physical branch is frozen or updated with a much smaller learning rate to preserve the learned mechanical response skeleton.

4.3. *Performance comparison of different models*

To evaluate the engineering applicability of the proposed framework

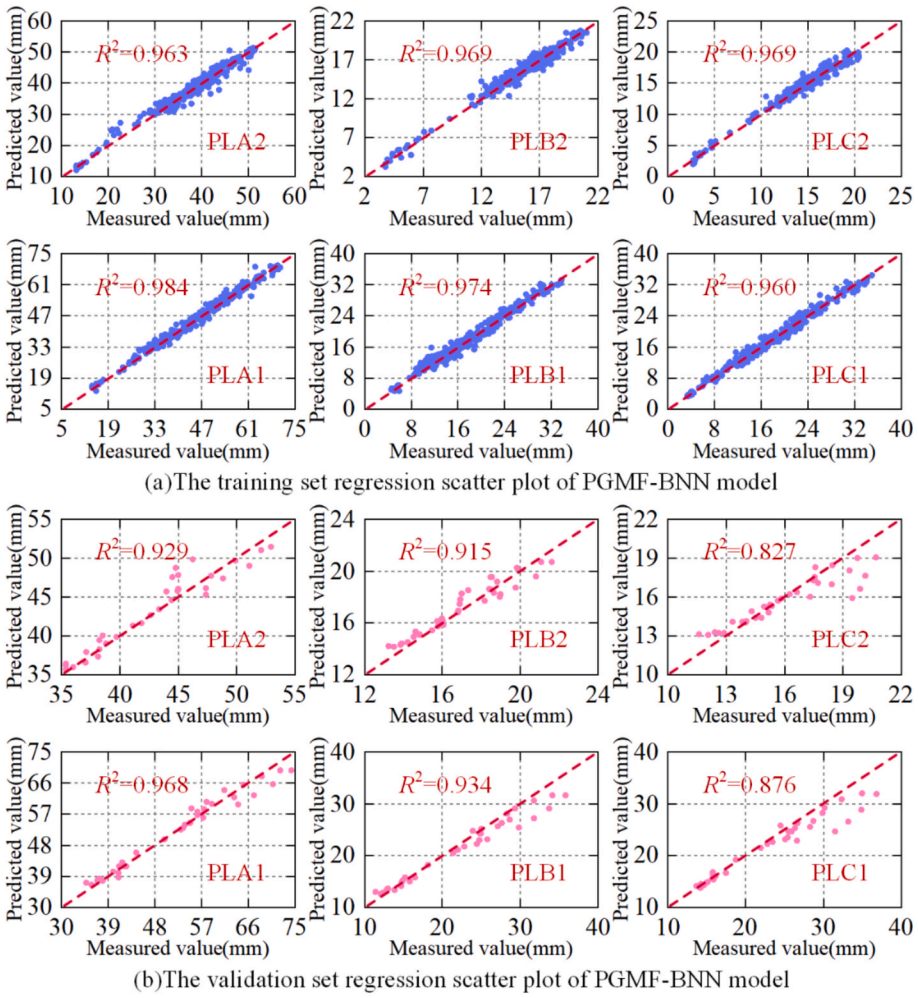


Fig. 8. Regression scatter plots of the training set and validation set for six representative monitoring points.

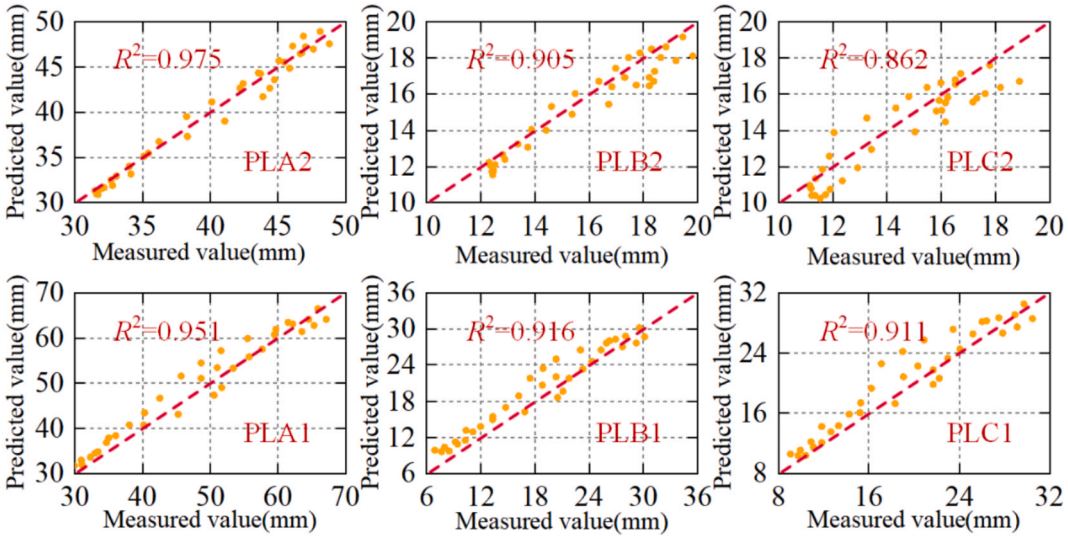


Fig. 9. Regression scatter plots of the testing set for six representative monitoring points.

under normal operating conditions, the physics-guided multi-fidelity Bayesian neural network (PGMF-BNN) was compared with several representative baseline models that reflect different modeling philosophies and levels of engineering prior incorporation. The purpose of this comparison is twofold, one is to verify the effectiveness of introducing physical priors and Bayesian inference into deformation prediction, further, to assess whether the proposed framework can provide a reliable modeling basis for dynamic state mapping in digital twin

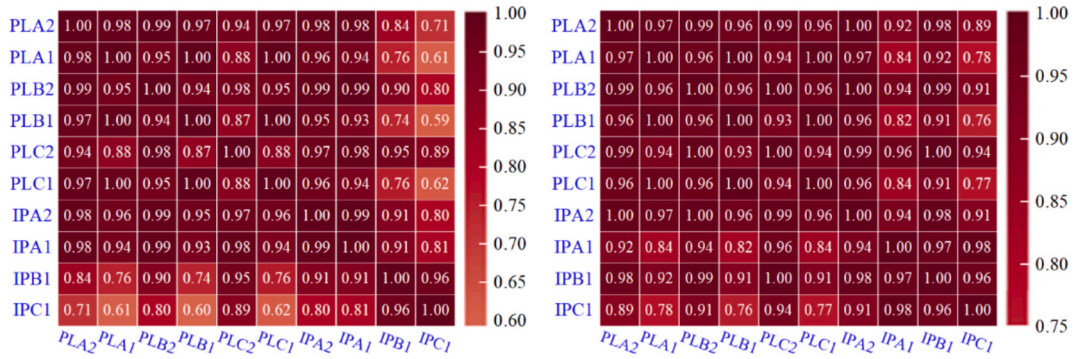


Fig. 10. Comparison between measured and reconstructed spatial correlation matrices of multi-point deformation.

Table 5
Computational cost of the proposed PGMF-BNN framework.

Category	Item	Value
Hardware	Device used by code	CPU
Hardware	CPU cores	20
Hardware	GPU	CPU only
Hardware	PyTorch version	2.8.0 + cpu
Training cost	Stage I physics-guided pre-training	6.818 s
Training cost	Stage II fine-tuning	6.364 s
Probabilistic inference cost	MC Dropout probabilistic inference	25.666 ms
Probabilistic inference cost	Average probabilistic inference time	0.733 ms/sample
Overall cost	Complete workflow	14.001 s

applications. The comparison models are summarized in Table 6.

For fair comparison, all models were evaluated under the same chronological data partition and testing set. The prediction targets were the same six representative plumb-line monitoring points, and the same evaluation metrics, including R^2, RMSE, MAE, and MAPE, were used for all models. The neural-network-based models were trained using the same training set, and their hyperparameters were determined based on validation-set performance. The testing set was used only for final evaluation. It should be noted that HF-only (Raw) and HF-only (Eng) were intentionally constructed with different input forms: HF-only (Raw) used only raw monitoring variables, while HF-only (Eng) used the same engineered variables as PGMF-BNN but without low-fidelity pre-training. This design was adopted to isolate the contribution of engineering feature construction and physics-guided multi-fidelity learning.

Fig. 11(a)–(f) and Fig. 12(a)–(f) presents the prediction results of different models were compared on both the training set and the validation set, respectively. Overall, the proposed model shows consistently superior fitting performance at all six representative monitoring points and is able to reproduce both the long-term trend and the local fluctuation characteristics of the measured displacement responses.

Table 6
Details of multiple comparison models.

Model category	The engineering characteristics of the model
Linear Regression	Represents a traditional statistical regression model, serving as the linear baseline
HF-only (Raw)	A standard neural network fed only with raw monitoring data, simulating a modeling scenario without engineering knowledge
HF-only (Eng)	Uses the same engineering features as PGMF-BNN but without physical pre-training, to isolate the respective contributions of feature engineering and physical guidance
LF-only	The physical branch trained solely using finite element simulation data, representing the output of a purely theoretical model
PGMF-BNN	The physics-guided multi-fidelity digital twin model proposed in this study

As shown in Fig. 11, PGMF-BNN, HF-only (Eng), and Linear can all reproduce the overall temporal evolution of the monitoring data in the training set, indicating that the adopted variables, including water level, temperature-related features, and time-effect factors, provide an effective basis for deformation prediction. In contrast, HF-only (Raw) fails to capture the temporal variation pattern, whereas LF-only shows systematic bias in displacement magnitude, highlighting the limitations of using raw monitoring inputs alone or relying solely on finite element simulation outputs. A closer examination reveals clear differences in reproducing peak responses. At PLA2 and PLA1, the Linear model follows the overall trend but deviates noticeably near several peaks, suggesting that conventional linear models are inadequate for capturing nonlinear deformation under unfavorable hydraulic and thermal conditions. By comparison, PGMF-BNN remains closely aligned with the measured curves, especially at peak positions, indicating a stronger capability to characterize the nonlinear relationship between external loading and structural response.

Although HF-only (Eng) also achieves relatively good accuracy, its predictions exhibit stronger local oscillations and greater sensitivity to measurement fluctuations, particularly near sharp changes and local extrema. This is more evident in the enlarged views of Fig. 12, where PGMF-BNN produces smoother curves and more accurate peak amplitudes. The result suggests that, even with the same engineered inputs, purely data-driven learning is still more susceptible to noise. By incorporating physics-guided pre-training, PGMF-BNN achieves a better balance between predictive accuracy and physical consistency. The deficiencies of LF-only and HF-only (Raw) are also evident in both figures. LF-only can reproduce the general cyclic trend but consistently underestimates the displacement amplitude, indicating that low-fidelity simulation data alone can only provide a coarse mechanical approximation. HF-only (Raw) fails to match both the displacement level and the fluctuation pattern, further confirming that direct regression on raw monitoring variables is inadequate for this strongly coupled nonlinear problem.

Overall, Figs. 11 and 12 illustrate that PGMF-BNN outperforms the comparison models in terms of prediction accuracy, temporal stability, and peak-response tracking, demonstrating the effectiveness of combining physics-guided pre-training with multi-fidelity learning for dam deformation prediction under normal operating conditions.

To quantitatively evaluate model performance, the coefficient of determination (R^2) and the root mean square error (*RMSE*) were selected as the primary evaluation metrics. In addition, the mean absolute error (*MAE*) and the mean absolute percentage error (*MAPE*) were supplemented to enable a more comprehensive comparison of the predictive accuracy. The predictive performance of the compared models at the six representative monitoring points is summarized in Table 7.

Considering the randomness involved in neural-network initialization and MC Dropout-based probabilistic inference, an additional stability analysis was conducted for the proposed PGMF-BNN. The model

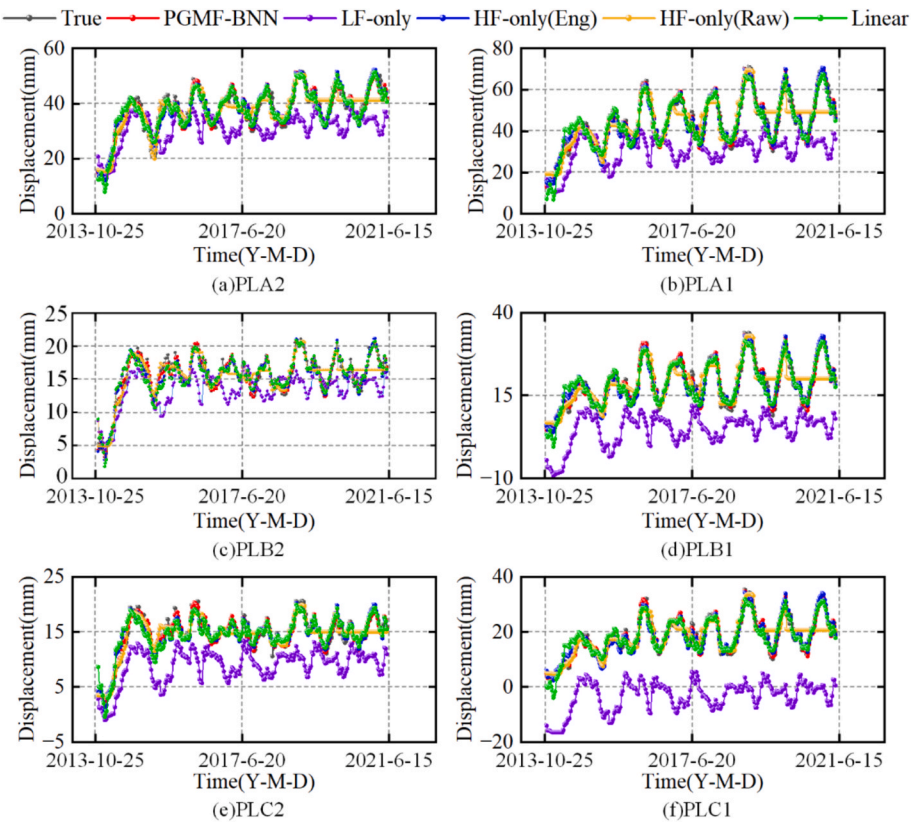


Fig. 11. Comparison of prediction performance of different models in the training set of six key monitoring points.

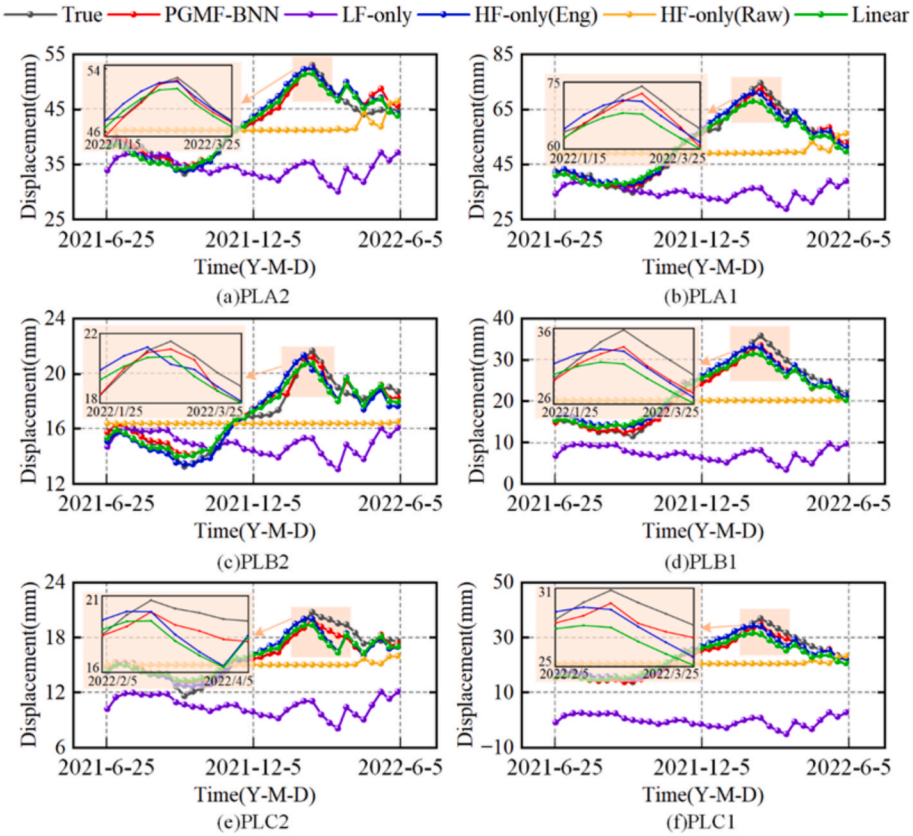


Fig. 12. Comparison of prediction performance of different models in the validation set of six key monitoring points.

Table 7
Comparison of the predictive performance of different models at key monitoring points.

Model	Index	PLA2	PLA1	PLB2	PLB1	PLC2	PLC1
PGMF-BNN	R^2	**0.975**	**0.965**	**0.924**	**0.951**	**0.885**	**0.932**
	RMSE(mm)	**0.934**	**1.169**	**0.725**	**1.602**	1.074	**1.733**
	MAE(mm)	**0.774**	**1.757**	**0.541**	**1.244**	**0.849**	**1.286**
	MAPE(%)	**1.882**	**3.639**	**3.236**	**5.634**	**5.935**	**6.803**
Linear	R^2	0.964	0.906	0.904	0.836	0.819	0.822
	RMSE(mm)	1.134	3.715	0.815	2.919	1.031	2.798
	MAE(mm)	0.972	3.177	0.897	2.462	2.294	1.785
	MAPE(%)	2.126	7.309	3.531	18.318	7.378	14.146
HF-only(Eng)	R^2	0.943	0.929	0.893	0.886	0.829	0.909
	RMSE(mm)	1.419	3.218	0.861	2.433	**1.005**	1.994
	MAE(mm)	1.197	2.785	0.746	2.106	1.048	1.586
	MAPE(%)	3.023	6.408	3.931	15.325	6.985	8.868
HF-only(Raw)	R^2	−13.054	−1.706	−22.851	−0.225	−0.483	−8.256
	RMSE(mm)	22.253	19.934	12.821	7.992	2.958	20.179
	MAE(mm)	21.811	18.342	11.992	6.436	2.572	19.157
	MAPE(%)	56.384	44.512	76.921	54.032	19.337	124.066
LF-only	R^2	−8.395	−2.043	−0.221	−0.205	−1.691	−0.265
	RMSE(mm)	18.20	21.144	2.911	7.924	3.984	7.458
	MAE(mm)	17.045	16.915	2.464	6.775	3.456	6.541
	MAPE(%)	40.969	30.861	15.109	44.951	26.519	40.756

was independently repeated five times using different random seeds, while the same chronological data partition, hyperparameter settings, and MC Dropout sampling number were maintained in all runs. The mean and standard deviation of the average performance over the six representative monitoring points are summarized in Table 8.

Results in Table 7 reflects that the proposed PGMF-BNN achieved the best and most stable performance among all competing models. In terms of predictive performance, the R^2 values of PGMF-BNN exceeded 0.90 at five of the six monitoring points, with particularly high values of 0.975 and 0.965 at PLA2 and PLA1, respectively. Even at PLC2, where the structural response is relatively more difficult to predict, the model still achieved an R^2 of 0.885, indicating satisfactory explanatory capability. From the perspective of absolute error, the RMSE values of PGMF-BNN remained consistently lower than those of the comparison models, while the corresponding MAE values also stayed at a relatively low level. In terms of relative error, the MAPE values of PGMF-BNN were all below 7%, demonstrating robust predictive accuracy across different monitoring locations.

By contrast, the traditional Linear Regression model showed noticeably weaker performance, especially at monitoring points with stronger nonlinear responses. For example, at PLB1, its MAPE reached 18.318%, which was substantially higher than that of PGMF-BNN. The performance gap became even more pronounced for the purely data-driven HF-only (Raw) model, whose R^2 values were negative at all monitoring points. These negative R^2 values indicate that the prediction errors of HF-only (Raw) exceed the inherent variance of the observed displacement data, meaning that the model performs worse than using the mean observed value as a constant baseline. Therefore, the negative R^2 values should be interpreted as evidence of severe underfitting rather than a calculation error. Its relative errors were also extremely large; notably, the MAPE at PLC1 reached 124.066%. These results indicate that direct use of raw monitoring variables is insufficient for reliable deformation prediction in this problem, owing to the strong nonlinearity and complex coupling among hydraulic, thermal, and time-dependent effects. A further comparison between HF-only (Eng) and HF-only (Raw) highlights the importance of physically meaningful feature construction. After introducing engineered variables derived from the deformation mechanism, the predictive performance improved markedly, with R^2 values generally exceeding 0.80 at most monitoring points. This result confirms that engineering features with clear physical semantics are a necessary prerequisite for effective data-driven modeling for dam deformation prediction.

Table 8
Stability analysis of PGMF-BNN under five repeated runs with different random seeds.

Metric	Mean ± Std
Average R^2 of six representative points	0.931 ± 0.009
Average RMSE of six representative points (mm)	1.243 ± 0.073
Average MAE of six representative points (mm)	1.031 ± 0.054
Average MAPE of six representative points (%)	4.735 ± 0.304

More importantly, even when using the same set of engineered inputs, PGMF-BNN still consistently outperformed HF-only (Eng), demonstrating the additional value of the proposed physics-guided multi-fidelity learning strategy. For instance, at PLA1, the RMSE was reduced from 3.218 mm for HF-only (Eng) to 1.169 mm for PGMF-BNN. Similar improvements can also be observed at the other monitoring points. These results indicate that the low-fidelity pre-training stage provides a mechanically informed initialization for the model, enabling it to better capture the underlying deformation mechanism and avoid suboptimal fitting driven solely by limited monitoring data. The results of the LF-only model further illustrate the limitation of relying exclusively on finite element simulation outputs. Although this model retains a certain degree of physical interpretability, its prediction accuracy is clearly inferior to that of PGMF-BNN, especially at monitoring points with more pronounced environmental and temporal effects. This finding suggests that finite element simulations alone can only provide a coarse mechanical response skeleton and cannot fully reproduce the actual structural state without correction from high-fidelity monitoring data.

To further quantify the spatial performance differences among the compared models, performance-gap heatmaps were constructed for the four evaluation metrics, as shown in Fig. 13(a)–(d). The gap was defined as the algebraic difference between each model and the best-performing model at the same monitoring point. Specifically, for the fitting metric, the gap is calculated as $\Delta R^2 = R^2_{\text{model}} - R^2_{best}$, where values are non-positive and a value closer to zero indicates better fitting performance. For the error metrics, the gaps are defined as $\Delta RMSE = RMSE_{model} - RMSE_{best}$, $\Delta MAE = MAE_{model} - MAE_{best}$, and $\Delta MAPE = MAPE_{model} - MAPE_{best}$, where values are non-negative and smaller values indicate closer performance to the optimum.

As shown in Fig. 13, the proposed PGMF-BNN consistently exhibits gap values close to zero at almost all monitoring points and across all metrics, indicating that it achieves the best overall predictive performance and the strongest spatial stability among the compared models. In contrast, the largest performance gaps are concentrated in the HF-only (Raw) model. For example, in Fig. 13(a), the R^2 gap at PLB2 reaches − 23.77, while in Fig. 13(d), the MAPE gap at PLC1 reaches 117.263. It

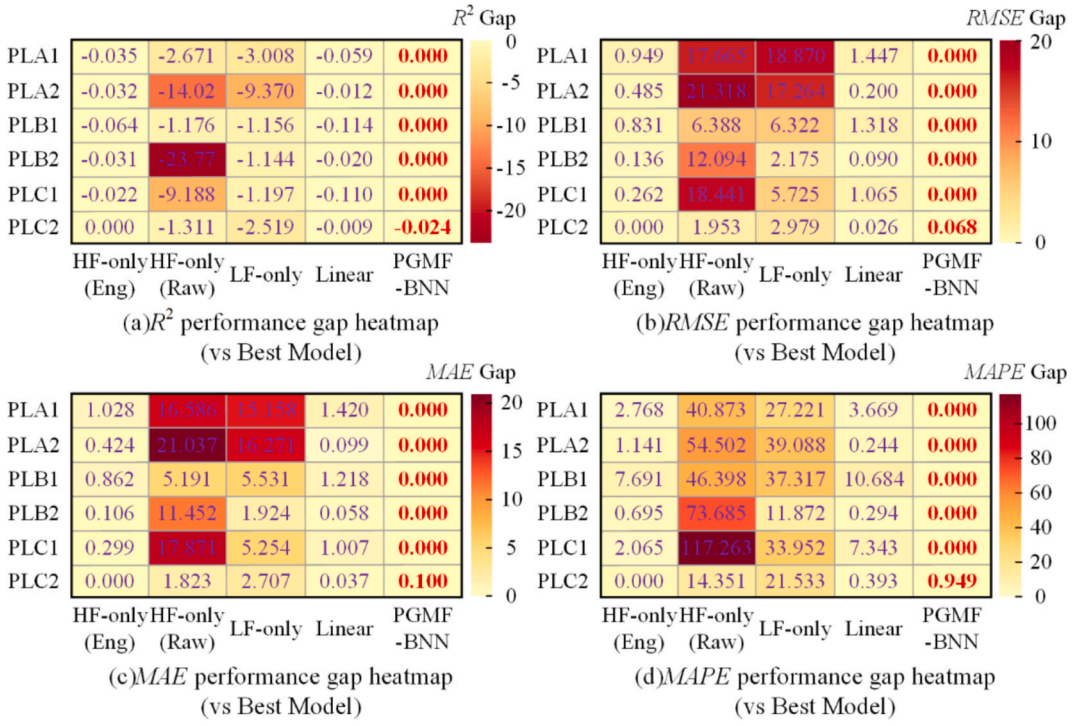


Fig. 13. Thermal map of spatial errors.

should be emphasized that the negative R^2 values of HF-only (Raw) reported in Table 7 do not indicate a calculation error. According to the definition of R^2, a negative value occurs when the prediction error variance exceeds the inherent variance of the observed data, meaning that the model performs worse than using the mean observed value as a constant prediction baseline. Therefore, the large negative R^2 values and the pronounced gap values of HF-only (Raw) jointly indicate severe underfitting rather than a normal prediction error fluctuation. These results demonstrate that directly using raw monitoring variables is insufficient for describing the highly coupled nonlinear deformation behavior of the arch dam.

Compared with HF-only (Raw), the Linear and HF-only (Eng) models show markedly smaller performance gaps, suggesting that engineering feature construction can substantially improve predictive capability. Nevertheless, noticeable discrepancies from PGMF-BNN remain at key monitoring points such as PLA1 and PLA2. For instance, in Fig. 13(b), the RMSE gap of HF-only (Eng) at PLA1 is still 0.949 mm, indicating that even with engineered inputs, purely data-driven learning remains limited in capturing the complex deformation mechanism without physics-guided pre-training. These results further confirm that the proposed physics-guided multi-fidelity strategy improves not only the overall prediction accuracy but also the spatial stability of multi-point deformation prediction.

As shown in Table 8, the standard deviations of all evaluation metrics are relatively small, with the standard deviation of the average R^2 being only 0.009 and that of the average RMSE being 0.073 mm. These results indicate that the proposed PGMF-BNN maintains stable predictive performance under different random seeds, and the reported results are not sensitive to random initialization or MC Dropout sampling.

In summary, PGMF-BNN has significant advantages in prediction accuracy, spatial consistency, and response stability. The above results further confirm the applicability of the model as an online state mapping digital twin prediction model for arch dams.

4.4. Ablation experiment

This experiment was conducted to clarify the role of the proposed two-stage transfer learning strategy in virtual–physical discrepancy correction. The optimization processes of Stage I and Stage II were evaluated using the loss functions L_{P1} and L_{P2}, respectively, and the corresponding deterministic real-time prediction trajectories are shown in Fig. 14. As shown in Fig. 14(a), the loss in Stage I decreases rapidly at the beginning of training and then stabilizes, indicating that the physical branch can efficiently learn the dominant mechanical response pattern from low-fidelity simulation data. In Stage II, the loss in Fig. 14(b) also converges smoothly without obvious oscillation, showing that the mechanically consistent initialization obtained in Stage I provides a stable starting point for high-fidelity discrepancy correction.

To mimic the online operation mode of a digital twin system, the testing samples were sequentially fed into the model in chronological order. The resulting real-time prediction trajectories at PLA2 and PLA1 are shown in Fig. 14(c) and Fig. 14(d). The Stage I predictions (orange) capture the general deformation trend but exhibit evident deviation from the measured data. After introducing the residual branch in Stage II, the corrected predictions (purple) become much closer to the measured responses and more accurately reproduce the temporal variation. These results indicate that the effectiveness of the proposed framework fundamentally relies on the two-stage transfer learning strategy, whereby Stage I establishes a mechanically consistent response skeleton and Stage II enables progressive correction of virtual–physical discrepancies under real operating conditions.

4.5. Physical consistency verification and response component decoupling

To enhance the interpretability of the proposed deep-learning framework, this section investigates the physical rationality of the internal mapping mechanism of PGMF-BNN through monotonicity analysis and component-wise decoupling. The aim is to examine whether the model preserves mechanically meaningful response characteristics while achieving accurate deformation prediction.

First, to assess whether the model remains consistent with fundamental mechanical principles under previously unseen operating conditions, a control-variable approach was employed to perform single-factor scanning of the input variables. As shown in Fig. 15(a), the

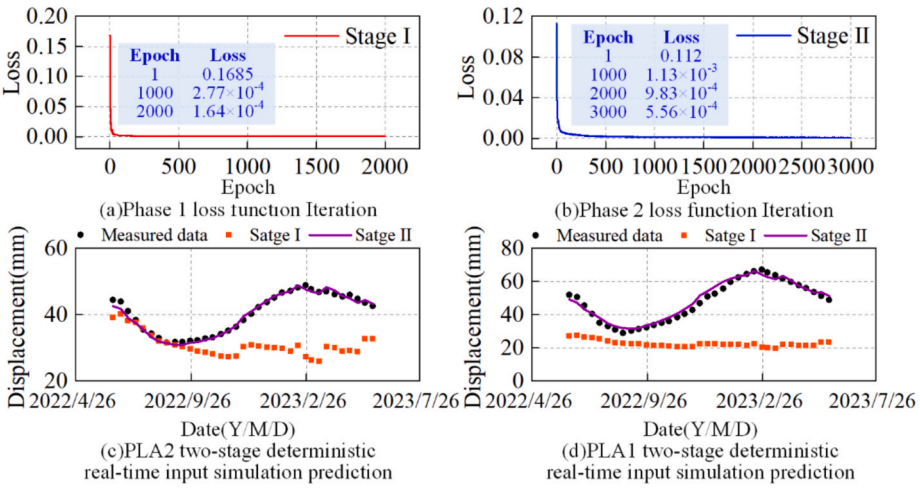


Fig. 14. Loss convergence curves and deterministic real-time prediction trajectories.

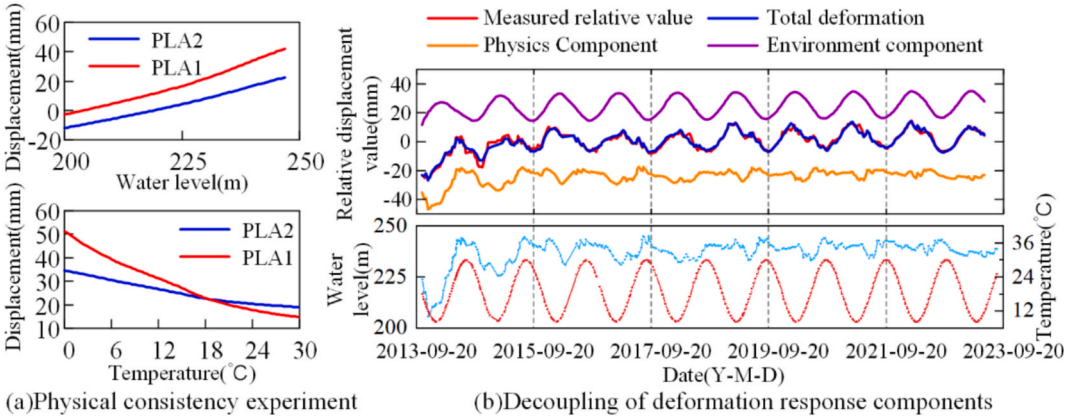


Fig. 15. Sensitivity analysis and physical decoupling of deformation components.

independent influence curves of water level and temperature on the crest displacement at PLA2 exhibit clear physically meaningful trends. With all other variables held constant, the predicted displacement increases monotonically with water level, revealing a positive nonlinear relationship between hydrostatic loading and radial displacement. This behavior is consistent with the basic mechanical principle that increasing water pressure leads to an increase in downstream radial deformation of the arch dam. By contrast, the predicted displacement decreases with increasing temperature, indicating a negative nonlinear correlation between temperature and deformation, which is also consistent with the thermal response characteristics commonly observed in arch dams. These results demonstrate that the proposed model is not merely a numerical fitting tool, but is capable of learning deformation-response relationships that are consistent with engineering mechanics.

Building upon this observation, a component-wise decoupling analysis was further conducted for the total deformation response in order to quantify the respective contributions of the physical branch and the residual branch. Considering that an inherent offset exists between the finite element simulation output and the reference level of the field monitoring data, a relative-displacement representation was adopted to facilitate comparison. As shown in Fig. 15(b), the measured relative displacement at PLA2 (red line), obtained after subtracting the mean value, fluctuates around zero. The output of the physical branch (orange line) exhibits a relatively stable periodic pattern primarily controlled by hydrostatic loading, and its temporal variation is broadly consistent with the water-level sequence shown in the lower panel of the figure. In contrast, the output of the residual branch (purple line) reflects the combined influence of temperature and time-dependent effects, and its overall variation is more closely aligned with the temperature sequence. A certain phase lag can also be observed, which is attributable to the time-dependent component incorporated in the residual branch.

After superposition of the two branch outputs, the reconstructed total deformation response (blue line) shows close agreement with the measured relative displacement. This result indicates that the proposed additive formulation can effectively decompose the overall prediction into physically interpretable components associated with dominant loading effects and environmental/time-dependent corrections. Consequently, the model not only provides accurate deformation prediction, but also offers an interpretable representation of the underlying response mechanism, thereby making it more suitable for digital twin-based monitoring, diagnosis, and engineering analysis.

4.6. Dynamic monitoring benchmark generation and probabilistic reliability assessment

Although the proposed PGMF-BNN has demonstrated high deterministic prediction accuracy in the preceding experiments, a digital twin model intended for risk-aware monitoring must further provide reliable probabilistic boundaries that can be dynamically updated with changing operating conditions. Based on the Bayesian inference framework, this study further evaluates the uncertainty-aware prediction capability of the model and constructs a dynamic monitoring benchmark for arch dam deformation.

Using the same recursive input strategy as in Section 4.2, the model

performs real-time probabilistic inference based on the current water level, temperature, and time-effect variables, and outputs time-varying prediction intervals accordingly. As shown in Fig. 16(a) and Fig. 16 (b), the probabilistic prediction results at monitoring points PLA2 and PLA1 exhibit a clear double-layer uncertainty structure that evolves continuously with time. In both cases, the measured data points are well enclosed by the outer uncertainty boundary throughout the testing period, indicating that the probabilistic benchmark constructed by the model can effectively characterize the fluctuation range of the actual structural response under in-service conditions.

The hierarchical uncertainty structure in Fig. 16 further provides an interpretable description of uncertainty sources. The inner yellow band represents the epistemic uncertainty, which is mainly associated with limitations in model parameters and sample support. On this basis, the outer black band incorporates the aleatoric uncertainty induced by environmental disturbances and observation noise, thereby forming the total predictive uncertainty envelope. This decomposition indicates that the proposed framework is capable not only of generating probabilistic prediction intervals, but also of distinguishing between uncertainty contributions arising from model insufficiency and those caused by external randomness.

From the perspective of digital twin applications, this result implies that the monitoring system is no longer restricted to fixed empirical thresholds or static warning intervals. Instead, it can generate adaptive health boundaries in real time according to the current operating conditions and the associated uncertainty level. Such a probabilistic benchmark is more consistent with the dynamic nature of dam behavior and therefore provides a more reliable basis for anomaly identification and risk early warning.

4.7. Reliability evaluation of monitoring data prediction under non-ideal operating conditions

To further evaluate the robustness of the proposed framework under non-ideal monitoring conditions, two types of robustness experiments were conducted, namely few-shot learning and noise anti-interference tests. In the few-shot learning experiment, only a given proportion of the high-fidelity training samples was used for Stage II fine-tuning, while the low-fidelity pre-training process and the testing set were kept unchanged. Specifically, the high-fidelity training pool was first defined as the first 80% of the chronological monitoring records, and subsets with proportions of 10%, 20%, and 50% were then selected from this training pool to simulate different levels of monitoring-data availability during early digital twin deployment. For each sample proportion, the model was first pre-trained using the same low-fidelity simulation dataset and then fine-tuned using the selected high-fidelity subset. The final performance was evaluated on the same testing set to ensure comparability among different few-shot settings.

In the noise-interference experiment, artificial Gaussian white noise was added to the normalized input variables rather than to the output displacement responses. Both the physical-branch inputs and the residual branch inputs were perturbed. The noise-contaminated input was defined as

$$\widetilde{x}_{i,j} = x_{i,j} + \rho\sigma_j\epsilon_{i,j}, \quad \epsilon_{i,j} \sim \mathcal{N}(0,1) \tag{25}$$

where $x_{i,j}$ denotes the normalized value of the j-th input variable of the i-th sample, $\widetilde{x}_{i,j}$ is the corresponding noise-contaminated input, σ_j is the standard deviation of the j-th input variable, and ρ denotes the random noise ratio. In this study, ρ was set to 0, 0.05, 0.10, 0.15, and 0.20, corresponding to noise levels of 0%, 5%, 10%, 15%, and 20%, respectively. This setting was used to examine whether the trained model could maintain stable prediction performance when the monitoring inputs were disturbed by different levels of random noise.

As shown in Figs. 17 and 18, the time-series prediction results provide a detailed illustration of model performance under few-shot learning and noise interference conditions. In the few-shot learning scenario (Fig. 17(a) and Fig. 17(b)), the model is able to capture the overall deformation trend even when only 10% of the high-fidelity samples are used, although noticeable deviations remain in peak amplitude and local variations. When the proportion of high-fidelity data increases to 20%, the agreement between predicted and measured curves improves significantly, particularly at peak and trough regions. With 50% of high-fidelity samples, the predicted trajectories almost coincide with the measured responses over most of the time period, indicating that the model can progressively refine prediction accuracy as more monitoring data become available.

Under noise interference conditions (Fig. 18(a) and Fig. 18(b)), the predicted curves remain stable and closely follow the measured data when the noise level is within 0%–10%, demonstrating strong robustness to moderate disturbances. As the noise intensity increases to 15% and 20%, the prediction results exhibit more pronounced local oscillations, especially near turning points and peak regions, although the overall deformation trend is still well preserved. This suggests that high noise levels mainly affect local fitting accuracy rather than the global response pattern.

The corresponding quantitative evaluation results are summarized in Fig. 19(a) and Fig. 19(b). In the few-shot learning case, both R^2 and *RMSE* show significant improvement with increasing high-fidelity sample proportion, which is consistent with the trend observed in the time-series curves. In the noise-interference case, the gradual decrease in R^2 and increase in *RMSE* with higher noise levels quantitatively confirm the degradation in local prediction accuracy observed in the time-domain results.

Collectively, the results in Fig. 19(a) and Fig. 19(b) demonstrate that the proposed PGMF-BNN maintains satisfactory predictive reliability

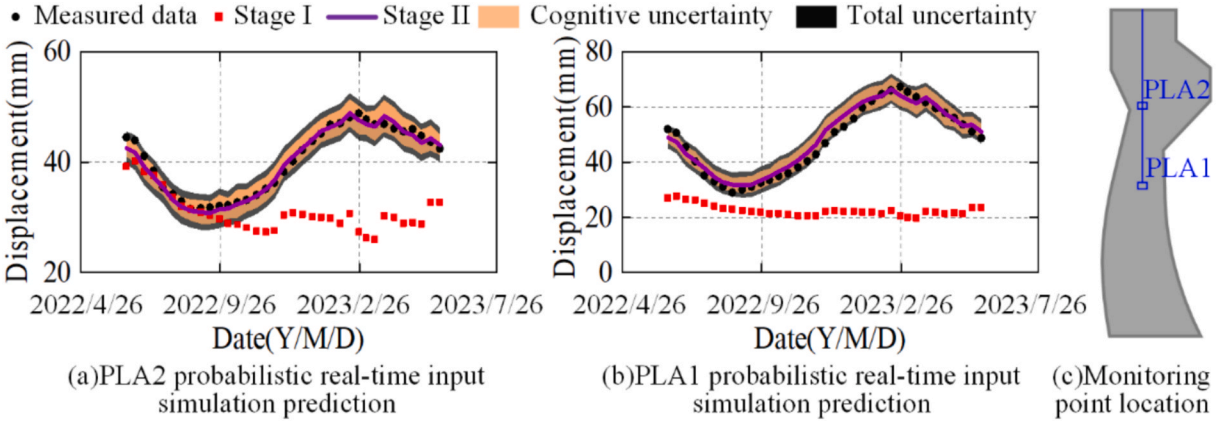


Fig. 16. Dynamic monitoring baseline and uncertainty decoupling via BNN.

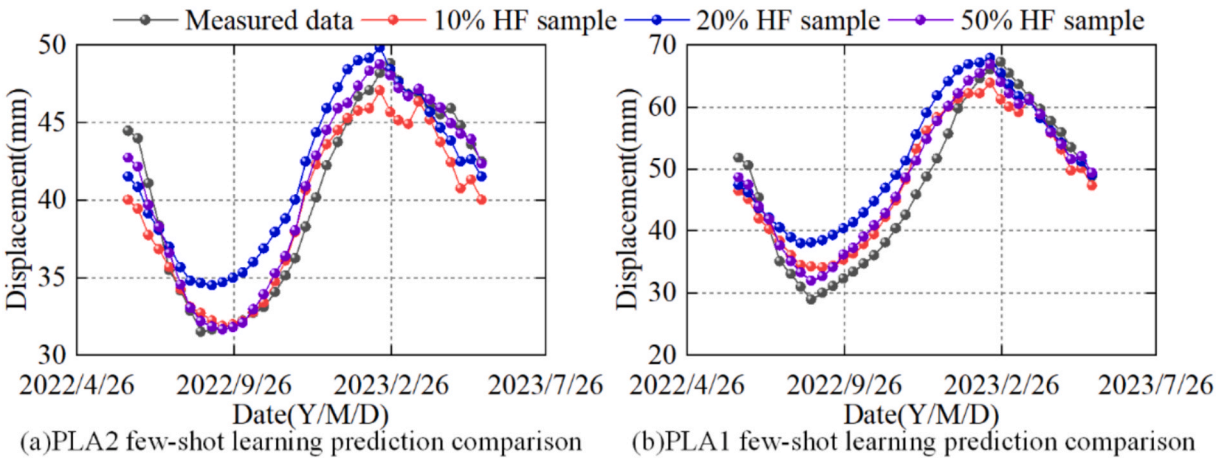


Fig. 17. Few-shot learning prediction performance under varying high-fidelity data proportions: (a)PLA2, (b) PLA1.

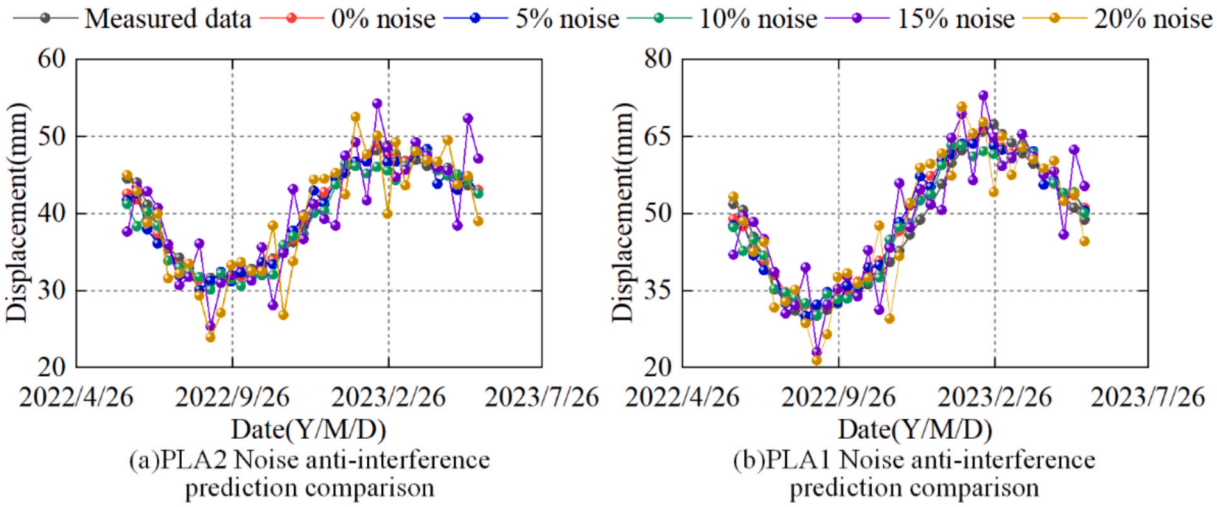


Fig. 18. Prediction performance under different levels of noise interference: (a)PLA2, (b) PLA1.

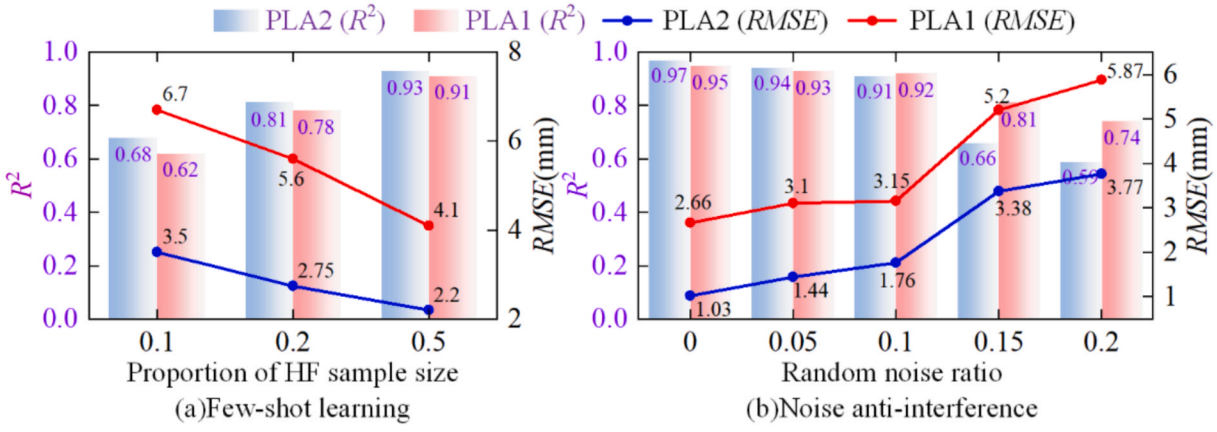


Fig. 19. Robustness evaluation under few-shot learning and noise interference.

under both data-limited and noise-contaminated conditions. This robustness can be attributed to the coordinated integration of the physics-based branch and the residual-correction branch, which enables the model to preserve mechanical consistency while retaining sufficient adaptability to non-ideal monitoring environments.

5. Conclusion

This study proposes a physics-guided multi-fidelity probabilistic prediction framework for concrete arch dams under digital twin-based monitoring. By integrating finite element simulations, monitoring data, spatial collaborative modeling, and Bayesian inference within a

unified framework, the proposed method addresses the key challenges of virtual–physical inconsistency, multi-point deformation coordination, and uncertainty-aware prediction. The main conclusions are as follows:

(1) A physics-guided multi-fidelity learning framework with MIMO-based collaborative modeling is developed to achieve both virtual–physical alignment and spatially coordinated real-time deformation prediction. Through a two-stage transfer learning strategy, the model first learns a mechanically consistent response skeleton from low-fidelity simulations and then progressively corrects the discrepancy using high-fidelity monitoring data. Meanwhile, the MIMO architecture enables the model to capture spatial correlations among monitoring points and reconstruct the coordinated deformation behavior of the dam. For the case-study dam, the proposed method achieved $R^2 > 0.90$ at five of the six representative monitoring points, with *MAPE* values below 7%.
(2) By incorporating Bayesian inference and MC Dropout, the framework extends deterministic prediction to probabilistic state characterization. The joint quantification of epistemic and aleatoric uncertainties enables the generation of calibrated prediction intervals and adaptive monitoring boundaries, providing a more realistic and interpretable basis for anomaly detection and risk early warning under evolving operating conditions.
(3) The robustness experiments under few-shot and noise-interference conditions demonstrate that the proposed framework maintains satisfactory predictive reliability in non-ideal monitoring scenarios. This confirms the effectiveness of integrating physical constraints, data-driven correction, and probabilistic inference, and highlights its practical applicability for digital twin-based safety monitoring of arch dams.

In summary, the proposed framework provides a physically consistent, spatially coordinated, and uncertainty-aware solution for dam deformation prediction, and offers a transferable methodological basis for digital twin applications in the safety monitoring and risk assessment of large hydraulic structures. Future studies will further validate the proposed framework in other hydraulic structures, such as gravity dams, embankment dams, and sluice structures. Although the finite element model, physical inputs, and monitoring layout should be adapted to specific structural characteristics, the core idea of multi-fidelity learning, virtual–physical correction, collaborative prediction, and uncertainty quantification remains transferable.

CRediT authorship contribution statement

Yingfan Chi: Investigation, Formal analysis, Data curation, Conceptualization. **Bowen Wei:** Validation, Funding acquisition. **Yating Hu:** Supervision, Investigation, Funding acquisition. **Dongyang Yuan:** Resources, Funding acquisition. **Zhaoxing Lei:** Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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